Confirming What Matters: Certified Third Normal Form Synthesis from Uncertain Functional Dependencies

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ABSTRACT

State-of-the-art schema design synthesizes a lossless, dependency-preserving Third Normal Form (3NF) decomposition that is optimal for given functional dependencies (FDs). Yet the FDs are estimated by mining data or asking large language models (LLMs), most mined FDs hold only by accident, and the models err. Since a single false FD can make the design lossy for the true FDs, a designer has to confirm the estimate, but cannot be asked about thousands of candidates. We propose to certify the design by a proof whose premises are the designer's answers and the data. The designer confirms a cover of the FDs that the design relies on, and refutes the maximal sets of attributes from which this cover does not derive a given attribute. We prove that the price of this certificate, one answer per FD of a smallest cover and one per such set that the data do not refute, is the length of a shortest proof. Our algorithm Cert3NF lets an advisor, built from the statistics of the data alone, choose the next question after every answer, so that a wrong advisor costs answers but never correctness. An evaluation by rules fixed in advance on 23 unseen relations shows that no fully automatic method we tested reliably obtains a good design, while a simulated designer certifies every design, without a budget, at about 1.3× to 1.4× the price in the median. Every certificate is correct where the data satisfy the true FDs, and the design is lossless for them even where a budget runs out. On synthetic schemata with up to a million candidates, the answers follow the price rather than the number of candidates. Answering every premise twice cuts the wrong certificates of a designer who errs at random from more than one half to about 2%, and in a study with nine people the experts certified all their designs. In summary, a designer's answers follow the length of the proof, however many FDs hold by accident.

PVLDB Reference Format:
Zhuoxing Zhang and Sebastian Link. Confirming What Matters: Certified Third Normal Form Synthesis from Uncertain Functional Dependencies. PVLDB, 20(1): XXX-XXX, 2027.
doi:XX.XX/XXX.XX

PVLDB Artifact Availability:
The source code, data, and/or other artifacts have been made available at https://github.com/zhuoxingzhang/cert3nf.

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Proceedings of the VLDB Endowment, Vol. 20, No. 1 ISSN 2150-8097.
doi:XX.XX/XXX.XX

1 INTRODUCTION

The logical design of relational database schemata is a classical topic of database research [12, 34]. Third Normal Form (3NF) synthesis turns a set of functional dependencies (FDs) into a lossless, dependency-preserving decomposition into 3NF [7, 10]. Recent algorithms also optimize the result, for example by minimizing the numbers of minimal keys and of non-key FDs that the schemata must maintain [51–53]. All of these guarantees hold relative to the input FDs.

Yet all approaches so far assume that the input is exactly the set Σ of business rules of the domain. In practice, Σ is estimated. FD discovery returns every minimal FD that holds on an instance [23, 39], and most of these hold by accident [47]. On the eight tables of TPC-H [43], for example, discovery returns 3,945 minimal FDs, of which only 76 follow from the specification. Large language models (LLMs) that judge FDs from the names of the attributes err frequently on real schemata. Hence, an estimate $\hat{\Sigma}$ differs from Σ, and synthesis computes the optimal design for the wrong input. A designer who knows the domain can tell genuine rules from accidental FDs, but cannot be asked about thousands of candidates. This is illustrated as follows.

Example 1.1 (Running example). Consider the schema Orders with the attributes O(rder), customer N(umber), C(ity) and D(ate). Its business rules are $\Sigma = \{O \to N, O \to D, N \to C\}$: an order has one customer and one date, and a customer lives in one city. Orders violates 3NF under these rules, since $N \to C$ holds while N is not a key, and 3NF synthesis returns the decomposition $\{ON, OD, NC\}$. On the following sample, FD discovery also reports $N \to D$, since every customer ordered on one date only.

Order	Cust. No.	City	Date
o_1	101	Auckland	3 March
o_2	101	Auckland	3 March
o_3	102	Wellington	3 March
o_4	103	Auckland	5 March
o_5	104	Wellington	3 March

On the estimate $\hat{\Sigma} = \Sigma \cup \{N \to D\}$, synthesis returns $\hat{\mathcal{D}} = \{ON, NC, ND\}$, which declares N as the key of ND. This design rejects legal data, such as a second order of customer 101 on another date. Synthesis adds no key schema, since ON contains the key O, and the design is lossless, dependency-preserving and in 3NF for $\hat{\Sigma}$, as synthesis guarantees for any input. Yet it is lossy for Σ: joining ON with ND assigns to every order of a customer the dates of all orders of this customer. In contrast, if the estimate misses $O \to D$, synthesis still returns $\{ON, NC, OD\}$, now with the key schema OD, which does not declare O as a key. Declaring it repairs the design without moving any data. □

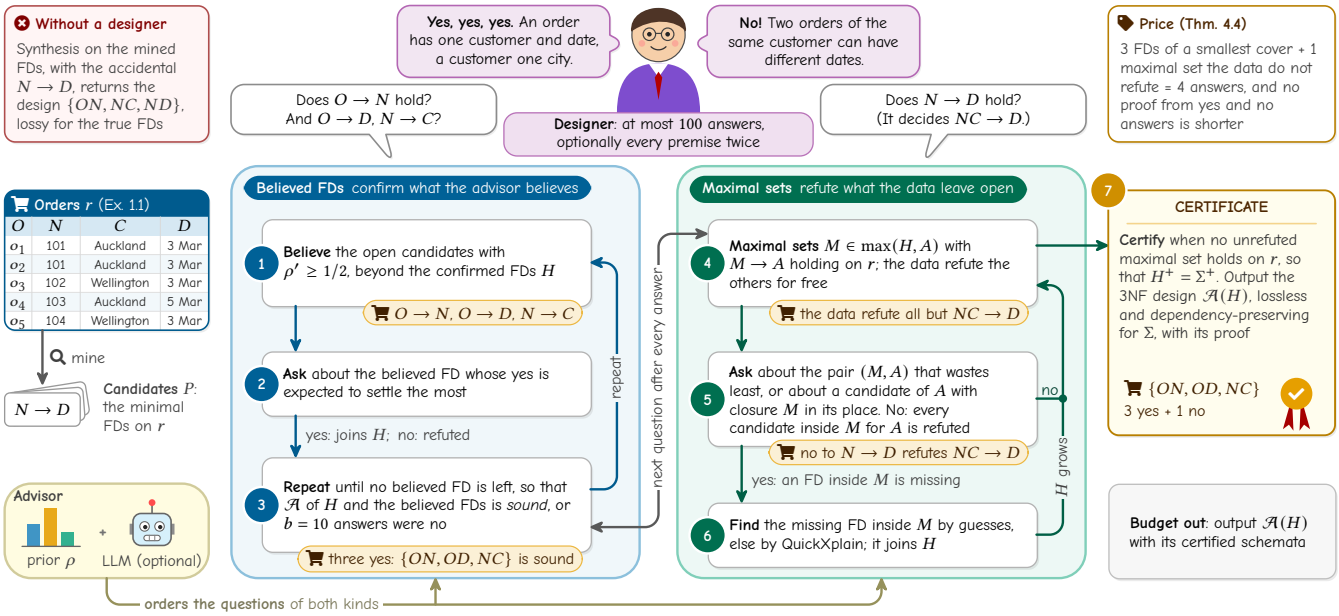


Figure 1: The Seven Steps of Cert3NF: Believed FDs and Maximal Sets, Chosen after Every Answer

Errors thus differ in their harm, and so does what a designer must confirm. Firstly, the false FD $N \to D$ broke losslessness for Σ, so its repair moves data, while the missing FD $O \to D$ only left a key undeclared, which a declaration repairs after deployment. Hence, we ask (Q1) *Which errors of an estimate must be resolved before deployment?*

Secondly, once the designer has confirmed the three rules of Σ, the design $\{ON, OD, NC\}$ is lossless whatever else holds, but whether it preserves every rule depends on what else holds, such as $N \to D$. The sample refutes every further FD except $N \to D$ and $NC \to D$, so whether any further rule holds is one question: can two orders of the same customer have different dates? If they can, the three confirmed rules are all there is. This leads to (Q2) *Which answers certify a design, and how many are needed?*

Thirdly, an LLM that reads the attribute names can answer that a customer does not determine a date, which is right here, but its answers are wrong on many other candidates. This raises (Q3) *What can advice that may be wrong contribute to a certificate?*

Finally, designers err, and a single wrong answer can certify a wrong design. We therefore ask (Q4) *Does a certificate survive a designer who errs?*

We answer these questions for the family of synthesis algorithms, including classical 3NF synthesis and iCONF [53], that eliminate redundant FDs from the atomic closure of their input. Our main idea is to certify the design by a proof whose premises are the designer's answers and the records of the data. This certificate shows that the confirmed FDs imply exactly the true FDs, so that the guarantees of synthesis for its input become guarantees for the true FDs. It has two parts. The designer confirms a cover of the FDs that the design relies on, which makes it lossless for the true FDs. The designer also refutes every *maximal set* of the confirmed FDs, which makes it dependency-preserving and in 3NF for them as well. Such a set is maximal among the sets of attributes from which the confirmed FDs do not derive a given attribute. Refuting it is the kind of answer that Armstrong relations and design by example ask for [6, 36]: two records can agree on these attributes and differ on that one. One such answer settles every candidate for that attribute whose left-hand side lies inside the set, and the data refute some of these sets for free. On the join of the lineitem and orders tables of TPC-H, for example, a certificate needs only 49 answers among 36,883 candidates, and our algorithm finds one with 54. Since its premises are answers and records, no estimate can invalidate the certificate.

Fig. 1 shows the resulting algorithm Cert3NF. It mines the candidates and attaches to each a probability from the statistics of the data, a source that we call the *advisor*. After every answer, Cert3NF asks the designer to confirm an FD that the advisor believes or, once none is left, to refute the maximal set that wastes the fewest answers in expectation, and it synthesizes the certified design from the confirmed FDs. That is, the advisor decides the order of the questions, and the designer decides the design. A designer who may err answers every premise of the proof a second time. Our **contributions** are as follows.

(1) Conceptually, we grade the damage of an estimate by the cost of its repair (Sec. 3, Q1). Only false FDs can make a design lossy for the true FDs, and a single one does so in exactly two ways, by fabricating a key or by short-circuiting true FDs, while missing FDs always admit illegal database states. Every candidate decides the constraints that the design declares, so a certificate has to settle every candidate, and it can only afford to do so in groups.
(2) Theoretically, we show that a confirmed cover and refuted maximal sets certify a design exactly (Sec. 4, Q2). The *price* of the certificate is one answer per FD of a smallest cover and

one per maximal set of the true FDs that the data do not refute, however many FDs hold by accident. No proof of the true FDs from yes and no answers is shorter.

(3) Algorithmically, we propose Cert3NF (Fig. 1, Sec. 5), in which an advisor built from the statistics of the data chooses the next question after every answer. Beyond one answer per refuted maximal set and per confirmed FD, a wrong advisor costs at most ten no answers, one wrong guess per refuted set and a logarithmic number of answers per FD it fails to predict, and never the correctness of the certificate (Q3). No method that does not know the true FDs avoids the logarithmic term. On data that violate true FDs in a few records, a tolerance keeps the certificate exact, since the data then refute only the FDs that fail on more than a given share of the records (Q2). For a designer who errs, every premise of the proof is answered twice, which bounds the probability of a wrong certificate (Q4).

(4) Empirically, we evaluate Cert3NF by rules fixed before any result was seen, on 23 confirmatory relations never used during development (Sec. 6). No fully automatic method that we tested, including an open LLM alone and a multi-agent normalizer, reliably obtains a good design on these relations. On them, a simulated designer certifies every design, without a budget, at about 1.3× to 1.4× the price in the median, and 19 to 22 of the 23 within a hundred answers, depending on the ground truth. Attribute exploration, the closest classical method, needs about 7× to 9× as many answers without a budget. EC^2, the strongest interactive baseline, certifies few designs, and as the order of the questions of Cert3NF it still spends 2× to 3× our answers above the price and certifies half as many designs within fifty answers. Every certificate is correct where the data satisfy the true FDs up to the tolerance (Q2). On synthetic joins with up to a million candidates, the answers follow the price rather than the number of candidates. Where the budget runs out, the design of Cert3NF is still lossless for the true FDs, while the interactive baselines, which accept unconfirmed FDs, return lossy designs in many such runs. The statistics of the data alone provide what the advisor needs, and the two LLMs we tested add at most one certified good design within a hundred answers and never invalidate a certificate, so the model stays optional (Q3). Answering every premise twice cuts the wrong certificates of a designer who errs independently from more than one half to about 2%, within a budget of a hundred answers. In a pre-registered study with nine people, the experts certified all their designs, in 26 minutes per session on average, while the errors of the practitioners show that a certificate is as correct as the beliefs of its designer (Q4).

(5) Practically, Cert3NF runs online at *https://cert3nf.onrender.com* for practitioners and researchers to design their own tables.

In summary, we establish the first approach to schema design that prices the certificate of a design against the unknown true FDs, lets the data refute the maximal sets that fail on them, and lets an unreliable advisor order the questions but never touch the correctness of the certificate.

Organization. Sec. 2 recalls 3NF synthesis, Sec. 3 grades errors by the cost of their repair, Sec. 4 defines the certificate and proves its price, Sec. 5 develops Cert3NF, and Sec. 6 presents our experiments. We discuss related work in Sec. 7 and conclude in Sec. 8. Appendix A contains the proofs and Appendix B the details of the experiments. Both are also in the full version of this paper, `Cert3NF-full.pdf`, in the repository of our code, online demo and results at *https://github.com/zhuoxingzhang/cert3nf*.

2 FUNDAMENTALS AND PREVIOUS WORK

We recall the notions of schema design to which the guarantees of synthesis refer, and introduce the elimination skeleton that the algorithms we study share. Ex. 1.1 illustrates them.

Dependencies. A relation schema is a finite set R of attributes. A functional dependency (FD) over R is an expression $X \to A$ with $X \subseteq R$ and $A \in R$, which a relation over R satisfies if any two of its tuples that agree on X also agree on A. For a set Σ of FDs, Σ^+ denotes the set of FDs that Σ implies [4], and $X^+_\Sigma = \{A : X \to A \in \Sigma^+\}$ the closure of X. The *atomic closure* Σ_a, which depends only on Σ^+, consists of the FDs $X \to A \in \Sigma^+$ with $A \notin X$ such that $Y \to A \notin \Sigma^+$ for every $Y \subsetneq X$. For $S \subseteq R$, the projection $\Sigma[S] = \{X \to A \in \Sigma_a : XA \subseteq S\}$ is a cover of the FDs in Σ^+ whose attributes lie within S. A key of (R, Σ) is a minimal $K \subseteq R$ with $K^+_\Sigma = R$, a superkey is a superset of a key, and an attribute is prime if it belongs to some key. A schema S is in 3NF with respect to Σ if, for every $X \to A \in \Sigma[S]$, X is a superkey of $(S, \Sigma[S])$ or A is prime for $(S, \Sigma[S])$. In Ex. 1.1, O is the only key of Orders under Σ, so $N \to C$ violates 3NF, and the projection of $\hat{\Sigma}$ onto ND is $\{N \to D\}$, while that of Σ is empty.

Designs. A design $\mathcal{D}$ over R is a decomposition of R, that is, a set of schemata whose union is R, and under a set Σ of FDs each schema $S \in \mathcal{D}$ declares the constraints $\Sigma[S]$. The design is lossless with respect to Σ if every relation over R that satisfies Σ is the natural join of its projections onto the schemata of $\mathcal{D}$, and it preserves Σ if $\bigcup_{S \in \mathcal{D}} \Sigma[S]$ implies Σ. We call $\mathcal{D}$ Σ-*valid* if it is lossless and dependency-preserving with respect to Σ and each of its schemata is in 3NF with respect to Σ. The following classical sufficient condition underlies 3NF synthesis.

Lemma 2.1 (Embedded cover [10]). *If a design $\mathcal{D}$ embeds a cover F of Σ, that is, each $X \to A \in F$ has $XA \subseteq S$ for some $S \in \mathcal{D}$, and some schema of $\mathcal{D}$ contains a key of (R, Σ), then $\mathcal{D}$ is lossless and dependency-preserving with respect to Σ.*

In Ex. 1.1, the design $\{ON, OD, NC\}$ embeds Σ and contains the key O, so it is lossless and dependency-preserving with respect to Σ.

The elimination skeleton. The synthesis algorithms we study differ only in the order in which they eliminate redundant FDs. This motivates the following definition.

Definition 2.2 (Elimination skeleton). Given Σ and an order π on Σ_a, the algorithm $\mathcal{A}_\pi$ starts with $\Sigma' = \Sigma_a$ and walks Σ_a in the order π. It removes each FD σ from Σ' that $\Sigma' \setminus \{\sigma\}$ implies, and keeps the schema XA of every FD $X \to A$ that it does not remove. The final Σ' is a non-redundant cover $G \subseteq \Sigma_a$ of Σ. The output $\mathcal{A}_\pi(\Sigma)$ consists of the maximal kept schemata, plus a minimum-size key of (R, Σ), chosen by a fixed order, if no kept schema contains a key.

That is, the skeleton keeps one schema per FD of a non-redundant cover, and the order decides which cover it keeps. Classical 3NF synthesis walks Σ_a by increasing size of the left-hand side and then

lexicographically in the order of the columns, and we use its variant that does not merge the schemata of FDs with equivalent left-hand sides. iCONF [53] parameterizes 3NF by the numbers of minimal keys and of non-key FDs of its schemata, and in the skeleton first walks the FDs whose schemata XA violate BCNF, by decreasing number of non-key FDs of XA, and then the others by decreasing number of minimal keys of XA. An FD that is walked earlier is more likely to be removed, so the order decides which schemata the result avoids. The next result shows that every order yields a valid design, and that the orders together reach every non-redundant cover within the atomic closure.

Lemma 2.3 (Skeleton guarantees). *Let π be computed from Σ_a, with ties broken by a fixed order of the attributes. Then $\mathcal{A}_\pi(\Sigma)$ depends only on Σ^+ and is Σ-valid. Over all orders, the covers G that $\mathcal{A}_\pi$ keeps are exactly the non-redundant covers of Σ contained in Σ_a.*

Hence, every member of the family inherits the guarantees of classical synthesis for the FD set it is given. What they mean for the true FDs when the given set is wrong, and how a designer can certify a design against the true FDs, have not been considered before. This is the contribution of our current work.

3 ERRORS AND THEIR DAMAGE

Synthesis runs on an estimate $\hat{\Sigma}$, while the domain imposes the true set Σ. By Lem. 2.3, the design is always $\hat{\Sigma}$-valid, so from now on, lossless, lossy, dependency-preserving and in 3NF without further qualification refer to Σ. We define the errors of $\hat{\Sigma}$ and their damage to the design, grade the damage by the cost of its repair, and show which errors a certificate must settle. We start with single errors.

Definition 3.1 (Single errors). An estimate $\hat{\Sigma} = \Sigma \cup \{\sigma\}$ with $\sigma \notin \Sigma^+$ has the single *false FD* σ. Given a non-redundant cover G of Σ, an estimate $\hat{\Sigma} = G \setminus \{\sigma\}$ with $\sigma \in G$ has the single *missing FD* σ.

In general, the false FDs of $\hat{\Sigma}$ are those in $\hat{\Sigma}^+ \setminus \Sigma^+$, and its missing FDs those in $\Sigma^+ \setminus \hat{\Sigma}^+$. In Ex. 1.1, $N \to D$ is a single false FD and $O \to D$ a single missing FD. Errors damage a design at two levels, the declared constraints and the schemata.

Definition 3.2 (Damage). Let $\hat{\mathcal{D}}$ be synthesized from $\hat{\Sigma}$, so that each schema S declares $\hat{\Sigma}[S]$. Relative to Σ, the design *over-enforces* if some declared FD is not in Σ^+, and *under-enforces locally* if $\Sigma[S] \not\subseteq (\hat{\Sigma}[S])^+$ for some schema S. It suffers *structural damage* if it is not Σ-valid.

The two levels differ in the cost of their repair. Over- and under-enforcement are repaired by declaring $\Sigma[S]$ instead of $\hat{\Sigma}[S]$ on the existing tables, which moves no data, while structural damage requires a new decomposition and a migration of the data. In Ex. 1.1, the design $\{ON, NC, ND\}$ over-enforces $N \to D$ and suffers structural damage, since it is lossy for Σ and does not preserve $O \to D$. In contrast, the design for the missing FD $O \to D$ is Σ-valid and only under-enforces locally. The direction of an error determines which damage at the level of constraints it causes, and only false FDs can cause lossiness, as the next result shows.

Proposition 3.3 (Asymmetry). *(a) If $\hat{\Sigma} \subseteq \Sigma^+$, every design that is lossless with respect to $\hat{\Sigma}$ is lossless with respect to Σ.*

(b) A design that preserves $\hat{\Sigma}$ over-enforces if and only if $\hat{\Sigma} \not\subseteq \Sigma^+$, and it then rejects a relation that satisfies Σ.

(c) If $\hat{\Sigma}^+ \subsetneq \Sigma^+$, every design that is lossless with respect to $\hat{\Sigma}$ accepts a database state that is not the projection of any relation satisfying Σ.

Since synthesized designs are $\hat{\Sigma}$-valid, Prop. 3.3 applies to them: a false FD always rejects legal data, while a missing FD never causes lossiness but always admits illegal states. In Ex. 1.1, the false FD $N \to D$ rejects a second order of customer 101 on another date, and the missing FD $O \to D$ admits two dates for one order.

Every candidate decides the constraints. Damage at the level of constraints is cheap to repair, but it cannot be ruled out without settling every candidate. We call two designs *constraint-equivalent* if they have the same schemata with equivalent declared constraints.

Proposition 3.4 (Constraint equivalence). *Let $\mathcal{A}$ be an algorithm of the family, and let $\mathcal{D}_1 = \mathcal{A}(\Sigma_1)$ and $\mathcal{D}_2 = \mathcal{A}(\Sigma_2)$ preserve Σ_1 and Σ_2, respectively. If $\mathcal{D}_1$ and $\mathcal{D}_2$ are constraint-equivalent, then ${\Sigma_1}^+ = {\Sigma_2}^+$.*

Every synthesized design preserves its input by Lem. 2.3, so two candidate sets that differ in their closure synthesize designs that differ in their constraints. Hence, whether a design declares the right constraints depends on every candidate, however unlikely. In Ex. 1.1, $O \to D$ only decides whether OD declares the key O, yet a certificate must still settle it. Sec. 4 shows that the answers of a designer settle the candidates in groups, so that the certificate stays affordable.

When a false FD breaks losslessness. Lossiness is the damage that moves data, and we characterize exactly when a single false FD causes it. For the rest of this section let $\sigma = X \to a$. Let $\Sigma[\hat{\mathcal{D}}] = \bigcup_{S \in \hat{\mathcal{D}}} \Sigma[S]$ be the true FDs embedded in $\hat{\mathcal{D}}$. The following theorem shows that lossiness arises in exactly two ways.

Theorem 3.5 (Lossiness). *Let $\hat{\Sigma} = \Sigma \cup \{\sigma\}$ have the single false FD σ, and let $\hat{\mathcal{D}} = \mathcal{A}_\pi(\hat{\Sigma})$ for any order π.*

(a) (Key loss) No schema of $\hat{\mathcal{D}}$ is a superkey with respect to Σ if and only if $a \notin S^+_\Sigma$ for every schema $S \in \hat{\mathcal{D}}$ with $S^+_{\hat{\Sigma}} = R$. In this case $\hat{\mathcal{D}}$ is lossy, and some minimal key of $(R, \hat{\Sigma})$ is not a superkey with respect to Σ.

(b) (Path loss) If some schema of $\hat{\mathcal{D}}$ is a superkey with respect to Σ but $\hat{\mathcal{D}}$ is lossy, then $\hat{\mathcal{D}}$ does not preserve Σ, and every FD $Y \to B \in \Sigma^+$ that $\Sigma[\hat{\mathcal{D}}]$ does not imply satisfies $X \subseteq Y^+_\Sigma$.

Under *key loss*, the design relies on a key that the false FD fabricated.

Example 3.6 (Key loss). Let $R = ABCD$ and $\Sigma = \{C \to D, BC \to A\}$, with the single key BC. The false FD $D \to C$ makes BD a key as well, and both classical synthesis and iCONF walk $BC \to A$ before $BD \to A$, remove it, and synthesize $\{ABD, CD\}$, whose key schema ABD contains BD. Since $(ABD)^+_\Sigma = ABD$ does not contain C, no schema is a superkey under Σ. Hence, the design is lossy. With the attributes in reverse order, both walk $BD \to A$ first and return $\{ABC, CD\}$, which is Σ-valid, so the order of the attributes decides whether this false FD breaks losslessness. □

Under *path loss*, the design contains a true superkey, but the true FDs that would carry it to every attribute have been short-circuited. Ex. 1.1 shows path loss. The false FD $N \to D$ makes $O \to D$ redundant, so synthesis keeps the cover $\{O \to N, N \to C, N \to$

$D\}$. The schema ON is a superkey under Σ, yet $\Sigma[\hat{\mathcal{D}}] = \{O \to N, N \to C\}$ carries it only to ONC. The missing attribute D is reached only through $O \to D$, which the false FD short-circuits, as $X = N \subseteq O^+_\Sigma$. Sec. 6 shows that nine in ten accidental FDs on TPC-H make the design lossy, all by path loss. In summary, a certificate has to establish that the design is lossless, which Thm. 3.5 ties to the FDs the design relies on, and that it declares the true constraints, which Prop. 3.4 ties to every candidate. The next section shows that the answers of a designer establish both.

4 CERTIFICATES FROM ANSWERS

We now define the certificate that Cert3NF issues, and show what it costs. The designer answers one kind of question, whether an FD $Z \to A$ holds in the domain, for any $Z \subseteq R$ and $A \in R$. Two records that agree on Z and differ on A are legal exactly when $Z \to A$ does not hold, so the question can also be put as whether such records can exist, with yes and no exchanged, as with Armstrong relations [6, 36].

Setting. The *known* FDs Σ_0 are true FDs given before any question, such as the FDs of a declared key, and Σ_0 may be empty. The candidates P are the minimal FDs that hold on the data and that Σ_0 does not imply, and we assume, as FD discovery does, that the data satisfy Σ. An FD that the data violate is then not a candidate and lies outside the certificate. The designer's answers divide into the FDs Σ_y answered yes and the FDs answered no, and $H = \Sigma_0 \cup \Sigma_y$ holds everything confirmed. A candidate is *open* if H does not imply it and no answer refutes it, where a no to $Z \to B$ refutes every FD $X \to A$ with $X \subseteq Z^+_H$ and $B \in (ZA)^+_H$, since $X \to A$ and H would imply $Z \to B$. We write $\mathcal{A}$ for the synthesis algorithm in use, and call a design $\mathcal{D}$ *good* for a set Σ' of FDs if it is Σ'-valid and $|\mathcal{D}| \le |\mathcal{A}(\Sigma')|$, that is, if it has every guarantee of synthesis for Σ' and no more schemata than synthesis needs.

Soundness and completeness. A certificate has to show that the design is good for Σ, which is unknown, from H, which is known. Confirmed FDs rule out false FDs, which gives soundness, and refuted maximal sets rule out missing FDs, which gives completeness. We assume for now that the designer answers truthfully, so $H \subseteq \Sigma^+$. Hence, a design that is good for H is lossless with respect to Σ by Prop. 3.3(a), and the tables it keeps are justified by confirmed FDs alone. We call such a design *sound*. What soundness leaves open is completeness: a true FD that H does not imply may be unpreserved, violate 3NF inside a schema, or admit a smaller design. Completeness is decided by the maximal sets of attributes from which H does not derive a given attribute. This motivates the following definition.

Definition 4.1 (Maximal sets). For a set H of FDs over R and an attribute $A \in R$, the *maximal sets* $\max(H, A)$ are the maximal $M \subseteq R$ with $A \notin M^+_H$.

Thus, H implies $X \to A$ if and only if X lies within no member of $\max(H, A)$, and the members of $\max(H, A)$ are the agree sets of an Armstrong relation for H that are maximal among those not containing A [6]. In Ex. 1.1, for $H = \Sigma$, the maximal set for D is NC: two orders of the same customer, who has one city, may still differ in the date, and so may any two orders that agree on less. The significance of these sets is embodied in the following theorem.

Theorem 4.2 (Certificate). *Let $H \subseteq \Sigma^+$, and let $\mathcal{D}$ be good for H. Then $H^+ = \Sigma^+$ if and only if $M \to A \notin \Sigma^+$ for every $A \in R$ and every $M \in \max(H, A)$. In this case, $\mathcal{D}$ is good for Σ, and $H[S] = \Sigma[S]$ for every schema $S \in \mathcal{D}$, so that every schema declares the true constraints.*

Hence, the guarantees of synthesis for H become guarantees for Σ. The equivalence rests on the classical fact that the maximal sets of a set of FDs determine its closure [6], so that each maximal set becomes one question to the designer. The certificate thus consists of yes answers that justify the design and no answers that refute the maximal sets that hold on the data, and no estimate enters it. Indeed, how the candidates were estimated, and which of them an LLM judged true, can decide which questions are asked, but not whether the answers prove the design good. We illustrate the theorem on our running example.

Example 4.3 (Certificate for Orders). Consider again Ex. 1.1 with $\Sigma_0 = \emptyset$. Once the designer has confirmed $O \to N$, $O \to D$ and $N \to C$, the design $\{ON, OD, NC\}$ is good for $H = \Sigma$, so it is sound. The maximal sets are NCD for O, CD for N, D for C and NC for D. The sample refutes three of them: o_1 and o_2 agree on NCD and differ on O, o_3 and o_5 agree on CD and differ on N, and o_1 and o_3 agree on D and differ on C. Only $NC \to D$ holds on the sample, and the designer answers that two orders of the same customer can have different dates. Hence, $H^+ = \Sigma^+$, and the design is certified by three yes answers and one no answer. □

The example shows that the data can refute maximal sets without a question, and that the remaining ones are exactly those that hold on the data. For a relation r that satisfies Σ, a set $M \subseteq R$ is *held* by r if $M \in \max(\Sigma, A)$ and $M \to A$ holds on r for some attribute A. We write m_r for the number of held sets, where a set that is maximal for several attributes counts once. Every agree set of r is closed under Σ, so the FD $M \to A$ with $M \in \max(\Sigma, A)$ fails on r exactly when M is the agree set of two records of r. A set is thus held for all attributes for which it is maximal or for none. Hence, m_r counts the maximal sets of Σ that are not agree sets of r, and it is zero exactly when r is an Armstrong relation for Σ [6]. A set of answers *certifies* a design if the design is good for H and every $M \in \max(H, A)$ with $M \to A$ holding on r is refuted by a no answer, so that the design is good for Σ by Thm. 4.2. The next result makes the example precise and gives the price of a certificate.

Theorem 4.4 (Price of a certificate). *Let r satisfy Σ, and let F be a set of FDs with $(\Sigma_0 \cup F)^+ = \Sigma^+$.*

- *(a) For every $H \supseteq \Sigma_0$, a maximal set $M \in \max(H, A)$ satisfies $M \to A$ on r if and only if M contains the left-hand side of a candidate $X \to A$.*
- *(b) Yes answers to the FDs of F and, for every held set M, a no answer to one FD $M \to A$ with $M \in \max(\Sigma, A)$ that holds on r certify every design that is good for Σ, with $|F| + m_r$ answers in all.*
- *(c) Every set of answers that certifies a design contains at least m_r no answers, and its yes answers Σ_y satisfy $(\Sigma_0 \cup \Sigma_y)^+ = \Sigma^+$.*
- *(d) A set of answers certifies a design that is good for Σ if and only if Σ^+ is the only closed set of FDs that holds on r, contains Σ_0 and the yes answers, and contains no FD answered no.*

We call $|F| + m_r$ for a smallest F the *price* of a certificate, and Thm. 4.4 shows that it is set by the domain and the data, not by the

number of candidates. Indeed, the yes answers are the cover that a designer would write down anyway, and the no answers are one per maximal set that the data do not refute, where one no settles at once every candidate that leads from inside the set to an attribute outside it. By (d), any answers that leave Σ^+ as the only closed set of FDs that agrees with them and holds on the data form a certificate, and by (c), these answers contain a set F and one no answer per held set. Hence, the price is the length of a shortest proof of the true FDs from yes and no answers. A smallest F is a minimum Horn base, which is NP-hard to find [19], but integer programming finds it in practice (Appendix B). FDs with sets of attributes on the right would have a smallest cover in polynomial time [34], but one yes would then confirm several FDs at once, while each of our questions asks about one attribute.

Part (a) also shows why the certificate refutes maximal sets rather than candidates. A true FD $Y \to A$ whose left-hand side is not minimal on the data is not a candidate, since some $X \subsetneq Y$ determines A on the data by accident. Refuting the candidate $X \to A$ would leave $Y \to A$ unsettled. In contrast, while H does not imply $Y \to A$, some $M \in \max(H, A)$ contains Y, so $M \to A$ holds on the data by (a) and is true, and the designer's yes to it reveals a missing FD inside M. The maximal sets can be exponentially many [6], but the price counts only the held ones. Since maximal sets are usually large, the question for M is best put by the attributes outside M: must two records that agree on everything but these also agree on A? Thm. 4.4 assumes that the right questions are asked, and the next section shows how to find them.

5 THE ALGORITHM CERT3NF

Cert3NF has to find the cover and the held sets of Σ without knowing Σ, within a budget of B answers, which we set to 100. It asks two kinds of question, shown in Fig. 1, and chooses the next one after every answer. The designer confirms the cover of the design that an *advisor* believes, and refutes the maximal sets of the confirmed FDs. The advisor is any source of a probability per candidate, and ours is built from the statistics of the data alone. It only orders the questions, and we bound what a wrong advisor can cost.

The advisor. An estimator assigns each candidate $c \in P$ a probability ρ_c from statistics of the data, such as the size of its left-hand side and the numbers of distinct values, by a logistic model trained on other schemata. Estimators trained elsewhere rank candidates well but are rarely calibrated on a new schema, so the advisor uses $\text{logit}^{-1}(\kappa \cdot \text{logit}\, \rho_c + \lambda)$ instead, with a prior on (κ, λ) learned on other schemata. The designer's answers about candidates update the posterior over (κ, λ). Candidates that share a left-hand side are not independent, since a left-hand side X that determines one attribute is likely to determine more. The advisor therefore treats every X as a determinant or not, and $X \to A$ can hold only if X is one. A yes on a candidate of X settles that X is a determinant and raises the other candidates of X to at least h, the rate at which a candidate holds given that another one of X holds, which is learned on other schemata and updated during the run. A no lowers the other candidates of X by Bayes' rule. The result is the probability ρ'_c of the advisor. An LLM may add evidence through the likelihood ratio of its answer, under error rates per schema with a prior learned on other schemata, and it is asked only where its answer can change the next question. Sec. 6.4 shows that the statistics alone do as well.

Believed FDs. The advisor *believes* the open candidates with $\rho'_c \geq 1/2$. These FDs and the confirmed FDs $H = \Sigma_0 \cup \Sigma_y$ form its estimate $\hat{\Sigma}$, and the *target design* is $\mathcal{A}(\hat{\Sigma})$. Cert3NF asks about the believed FD c with the largest $\rho'_c(1 + n_c)$, where n_c is the number of other believed FDs that $H \cup \{c\}$ implies, since its yes settles these without an answer of their own. A yes puts the FD into H, a no refutes it and every candidate that would imply it with H (Sec. 4), and the advisor and $\hat{\Sigma}$ are recomputed. Once no believed FD is left, $\hat{\Sigma}^+ = H^+$ and the target design is good for H and hence sound. Every yes answer puts a true FD into H, so an advisor that insists on false FDs costs at most b no answers, with $b = 10$, before Cert3NF stops asking about believed FDs. An advisor that believes more true FDs than a smallest cover needs costs yes answers instead, which Prop. 5.2 counts.

Maximal sets. When no believed FD is to be asked, Cert3NF turns to the maximal sets of H. By Thm. 4.4(a), a pair (M, A) with $M \in \max(H, A)$ needs a question only if M contains the left-hand side of a candidate of A; otherwise two records of r refute it without a question. A no answer to a held set is part of every proof, so only a yes can waste answers. Hence, Cert3NF asks about the pair with the smallest expected waste $p_M(1 + e_M)$, where $p_M = 1 - \prod_c (1 - \rho'_c)$ over the open candidates c of A inside M is the probability of a yes, and e_M is the expected number of answers that a yes costs beyond the one for its FD. A no refutes at once every candidate $X \to A'$ with $X \subseteq M$ and $A' \notin M$. A candidate $X \to A$ with $X^+_H = M$ is asked in place of $M \to A$, even before the believed FDs, since its no refutes $M \to A$ as well and its yes yields the FD with one answer instead of two.

Finding an FD. A yes to $M \to A$ means that an FD $X \to A$ with $X \subseteq M$ is missing from H, and Cert3NF finds a minimal one with at most $g = \lceil \log_2 |M| \rceil + 1$ tries. A true FD that is not minimal on the data has no candidate of its own, so Cert3NF first tries the left-hand sides inside M that the designer has confirmed for other attributes, while such tries keep hitting in this run. It then guesses, asking about the likeliest open candidate inside M while its probability given $M \to A$ is at least $\max(0.2, 1/g)$. A wrong guess is not wasted, since it refutes a candidate. If the tries miss, it asks about subsets of M by QuickXplain [25], with the attributes of M ordered by how much probability the open candidates place on them, and finds a left-hand side of k attributes with $O(k(1 + \log(|M|/k)))$ further answers. Where most maximal sets hold, even the yes is a waste, since a set whose first guess hits costs two answers and the guess alone costs one. With p the rate of maximal sets that held in this run and h' the rate of first guesses that hit, asking the guess first wastes $1 - ph'$ answers where the set first wastes p, so Cert3NF swaps them if and only if $p(1 + h') > 1$. After a miss, it asks about $M \to A$ at once. The FD found joins H, and the maximal sets of H are recomputed from the minimal keys of $H \cup \{A \to B : B \in R\}$ [33] by transversals. A run ends when no pair needs a question, that is, when every maximal set of H that holds on r is refuted. By Thm. 4.2, $H^+ = \Sigma^+$, and Cert3NF returns $\mathcal{A}(H)$ with its certificate: the confirmed FDs and the refuted pairs, each refuted by a no answer or by two records of r.

In Ex. 4.3, the statistics make O look like a key and the advisor doubts $N \to D$, so it believes $O \to N$, $O \to D$ and $N \to C$. The designer confirms them in three answers, and one more refutes the only maximal set that holds on the data, NC for D. Since $N_H^+ = NC$, Cert3NF asks about $N \to D$ in place of $NC \to D$, which is the question of the introduction. If the advisor also doubted $O \to D$, the designer would confirm $O \to N$ and $N \to C$, and Cert3NF would ask about $O \to D$ in place of the maximal set ONC for D, so that the certificate still takes four answers, the price.

Proposition 5.1 (Refuted maximal sets). *Let r satisfy Σ and the designer answer truthfully. Every no answer to a maximal set, or to a candidate asked in its place, falls on a held set, and no held set receives two. Hence, Cert3NF receives at most m_r such no answers.*

The next result adds the cap b, the first guesses and the questions that find each FD, and so counts all answers.

Proposition 5.2 (Number of answers). *Let r satisfy Σ and the designer answer truthfully, and let $J(k, n) \in O(k(1 + \log(n/k)))$ be the number of answers that QuickXplain needs to find a left-hand side of k attributes among n. Every yes answer to a candidate and every FD found through a maximal set is a different FD of Σ^+ that Σ_0 does not imply, with a left-hand side that is minimal under Σ. If there are d such FDs, Cert3NF asks at most*

$$b + m_r + \min(m_r, d) + d + \sum_{X \to A \text{ found by a yes to } (M,A)} \big(g + J(|X|, |M|)\big)$$

questions.

Hence, beyond the m_r no answers to maximal sets and one answer per FD, an advisor costs at most b no answers to believed FDs, g wrong tries per FD, and one wrong first guess per refuted set. The FDs that a yes reveals are found at a logarithmic cost, whatever the advisor says. On our relations, d is at most 1.4× the size of a smallest cover in the median, although it can be exponentially larger in general (Appendix A). The logarithmic term is the cost of not knowing Σ, and no method avoids it.

Proposition 5.3 (Lower bound). *For all $1 \le k < m$, let $R = MA$ with $|M| = m$ and $\Sigma_0 = \emptyset$. There is a relation r over R on which every $\Sigma = \{X \to A\}$ with $X \subseteq M$ and $|X| = k$ holds and has the price $k + 1$, while every method that certifies a design for each of them needs at least $\log_2 \binom{m}{k} \ge k \log_2(m/k)$ answers on one of them.*

Hence, no method certifies every design within a constant factor of the price. Without an advisor, Cert3NF needs at most $1 + k + J(k, m)$ answers on these relations, which is within a constant factor of the best method in the worst case, as every method also pays the price. An advisor that believes exactly $X \to A$ brings the cost down to the price.

Partial certificates. If the budget runs out, Cert3NF certifies what it can. A schema S of $\mathcal{A}(H)$ is certified on its own when every maximal set of the projection $H[S]$ is refuted, which leaves only the preservation of FDs across the schemata open.

Proposition 5.4 (Partial certificate). *Let $H \subseteq \Sigma^+$, let $S \subseteq R$, and let $T \to A \notin \Sigma^+$ for every $A \in S$ and every $T \in \max(H[S], A)$. Then Σ^+ and H^+ contain the same FDs over S, and a schema S of $\mathcal{A}(H)$ is in 3NF with respect to Σ and has $H[S] = \Sigma[S]$.*

Without a certificate, Cert3NF returns the sound design $\mathcal{A}(H)$ with its certified schemata. The design is lossless, its certified schemata already declare the true constraints, and a later session continues from its answers.

A designer who errs. The certificate of Thm. 4.2 is a proof, and a wrong answer is a false premise. The *premises* are a non-redundant subset of the yes answers that generates H with Σ_0, and a subset of the no answers that refutes every maximal set of H that holds on r. If all of them are right, so is the certificate. In its *careful mode*, Cert3NF therefore has the designer answer every premise a second time before it certifies, and a third time when the two answers disagree. The net answer of a question is the majority of its answers, and a question with one answer of each kind counts as unanswered. The set H holds the net yes answers only, so the designer is asked once more about the FD that QuickXplain finds. No question is asked more than three times, and the certificate is issued once every premise has a majority of at least two answers and no conflict is left (Appendix A).

Proposition 5.5 (Wrong certificates). *Let r satisfy Σ, let every answer be wrong independently with probability ε, and let Q be the expected number of different questions that Cert3NF asks in its careful mode. Then Cert3NF issues a wrong certificate with probability at most $Q(3\varepsilon^2 - 2\varepsilon^3)$, which is below $300\varepsilon^2$ within a budget of 100 answers.*

That is, if a designer errs on one answer in fifty, at least seven runs in eight end without a wrong certificate, on any data that satisfy Σ. The bound holds as well when every question has its own error rate ε_q, drawn independently from one distribution, with $3\mathrm{E}[\varepsilon_q^2] - 2\mathrm{E}[\varepsilon_q^3]$ in place of $3\varepsilon^2 - 2\varepsilon^3$. A designer who repeats an error has $\varepsilon_q \in \{0, 1\}$, for which this term equals ε, the error rate of a single answer.

Data that violate true FDs. The certificate also assumes that the data satisfy Σ, while real data violate true FDs in a few records. The data then refute such an FD without a question, so the designer never sees it. We say that $M \to A$ holds on *r up to θ* if at most a share θ of the records has to be removed for $M \to A$ to hold. This share is the measure g_3 of Kivinen and Mannila [27], which discovery of approximate FDs uses [23]. With a *tolerance* θ, Cert3NF asks the designer about every maximal set $M \in \max(H, A)$ for which $M \to A$ holds on r up to θ, and lets the data refute the others. The next result shows that the certificate then stays exact on data that satisfy the true FDs up to θ.

Proposition 5.6 (Tolerance). *Let every FD of Σ^+ hold on r up to θ, and read "holds on r" as "holds on r up to θ" and the candidates as the minimal FDs that hold on r up to θ and that Σ_0 does not imply. Then the data refute no FD of Σ^+, a set of answers that certifies a design proves it good for Σ, and Thm. 4.4(a)–(c) hold.*

Every maximal set that the data no longer refute costs a question, so a tolerance trades questions for robustness.

The posterior certificate. Instead of the maximal sets, a posterior over closed sets of candidates can certify a design by asking about the candidates of largest posterior marginal and accepting a sound design once it is good for a share $1 - \delta$ of the closed sets sampled from the posterior. We use this *posterior certificate* as a baseline. It rests on the error rates of the model and on the candidates. Sec. 6 shows that it is wrong where the model refutes a true

FD or a true FD is not minimal on the data, while the certificate of Thm. 4.2 cannot be.

6 EXPERIMENTS

Our experiments answer the empirical side of the four questions of the introduction, which build one argument. Single errors damage most synthesized designs, so designs must be certified (Q1). No automatic method reliably obtains a good design, so a designer pays the price of a certificate (Q2). An advisor lowers this cost (Q3), and certificates survive the slips of a designer (Q4).

6.1 Set up

Data. Random schemata and the eight tables of TPC-H [43] measure the damage of single errors (Q1). The certificates are evaluated on 59 relations with data and documented FDs, with 5 to 25 attributes and 200 to 2.7 million rows (Tab. 8 in Appendix B). Cert3NF was designed on the 36 *development* relations, and its prior is learned on them. The 23 *confirmatory* relations were chosen from their documentation alone by rules fixed in advance, and never opened during development. Seven of them are benchmarks whose schema a model may have memorized (class A), and sixteen are real data sets with a data dictionary (class B).

Ground truth. The ground truth of a relation consists of the FDs that its publisher documents, in a *full* version that includes the rules a reader has to infer and a *sure* version of the explicit ones. Together with the declared key as Σ_0, it plays the designer, and for (Q4) we inject wrong answers. A design is *good* if it is good for the documented FDs and the declared key (Sec. 4), and a certificate is *wrong* if it rests on a false FD or misses a documented FD that holds on the data. On five confirmatory relations, listed before any run and marked † in the tables, the data violate a documented FD of the full truth, which then lies outside every certificate.

Methods. We compare Cert3NF with three variants: *with LLM*, whose advisor also reads the answers of a model, *two phases*, which also reads them but confirms the believed cover first and then asks about the maximal sets, most likely first, and *without advisor*, which asks about the maximal sets alone, in a fixed order, with QuickXplain in place of the guesses. The interactive baselines are the *posterior certificate* of Sec. 5 and *EC*^{2}, which asks about the candidates whose answers separate the most pairs of posterior samples with different designs [18], both with the model. To separate our framework from the order of its questions, the variant *EC*^{2} *order* keeps the candidates, maximal sets and stopping rule of Cert3NF and chooses the next question by the score of EC^2, so that its certificate is exact as well. The fully automatic methods ask no designer: the mined FDs taken as true (*mined FDs*), the yes answers of the model (*trust*), a per-FD classifier on the statistics and the answer (*classifier*), the posterior alone (*posterior alone*), and the rules that the model states for the relation (*rules*). The model also designs the relation itself, when asked directly (*LLM design*) and as a multi-agent normalizer with a verifier after MARS [28] (*normalizer*). As a reference, *every candidate* lets the designer judge every candidate without a budget.

Protocol. The candidates are all minimal FDs of the data, mined by our exact implementation in the style of HyFD [39], and the LLM is Qwen3.5-9B. Cert3NF is deterministic, and every method with a random component runs with three seeds. Before any confirmatory result was seen, we fixed five rules V1 to V5 by which the claims are judged, per truth on all relations and on each class (Appendix B).

Table 1: Damage by a Single Error on Random Schemata with $n \leq 6$ Attributes (in %, $n = 7$ in Parentheses, – No Lossy Design)

	false FD		missing FD	
measure	classical	iCONF	classical	iCONF
over-enforcing	100 (100)	100 (100)	0 (0)	0 (0)
locally under-enforcing	0 (0)	0 (0)	73.1 (64.0)	73.0 (64.0)
lossy	51.1 (62.8)	50.8 (62.6)	0 (0)	0 (0)
not preserving	41.5 (28.0)	41.5 (27.1)	44.5 (59.2)	44.6 (59.2)
not in 3NF	10.4 (10.1)	8.5 (7.8)	30.8 (43.8)	30.7 (43.8)
structural damage	68.6 (73.5)	68.2 (72.8)	68.0 (87.5)	68.1 (87.5)
key loss, of lossy	63.6 (85.5)	64.1 (86.2)	–	–

Table 2: Damage by a Single Accidental FD on TPC-H (in %, iCONF in Parentheses Where It Differs)

table	$\|R\|$	accidental FDs	lossy	not in 3NF	structural damage
region	3	2	100	0	100
nation	4	3	100	0	100
partsupp	5	16	100	0	100
supplier	7	21	85.7	14.3 (4.8)	90.5
customer	8	34	94.1	0	94.1
orders	8	44	100	0	100
part	9	79	86.1	7.6 (3.8)	89.9
lineitem	16	3,670	89.3	0.6 (0.4)	89.8
all	–	3,869	89.5	0.8 (0.5)	90.0

6.2 Q1: Which Errors Damage the Design?

We first ask which errors damage a synthesized design, and how often.

Random schemata. Tab. 1 shows that one error causes structural damage to about two thirds of the designs for $n \leq 6$, and to up to 87.5% for $n = 7$, whichever algorithm synthesizes them. Indeed, every order walks the same wrong atomic closure with the same wrong keys and only selects among its covers. Hence, robustness to errors has to come from the estimate or from the designer, not from the choice of the synthesis algorithm. As Prop. 3.3 predicts (first five rows of Tab. 1), every false FD over-enforces and half of them make the design lossy, while missing FDs do neither and break preservation or 3NF instead.

TPC-H. Tab. 2 shows that accidental FDs damage the structure even more often than random false FDs: 90.0% of them cause structural damage, against 68.6% on random schemata. Indeed, in every table the primary key K determines each other attribute directly, so the left-hand side X of every accidental FD $X \to a$ lies in $K^+_\Sigma = R$, which is the condition of Thm. 3.5(b) for short-circuiting $K \to a$. Unless a true non-key FD also reaches a or leads from a into X, synthesis then stores a with X instead of with K, and the join on X, which is not a superkey under Σ, is lossy, as in Ex. 1.1. The two

Table 3: Answers until the Certificate on the Confirmatory Relations, with True Candidates and Price (Thm. 4.4) under the Full and the Sure Truth; in Parentheses the Answers without a Budget, Where They Exceed 100, – No Certificate, † a Design That Misses a Documented FD the Data Violate; LLM: Cert3NF with the LLM; EC² Ord.: Cert3NF with the Order of Questions of EC²; Post.: Posterior Certificate; Post. and EC² Ask about Candidates Alone; EC² Ord., Post. and EC²: Median of Three Seeds; $\theta = 0.1$: Cert3NF on the Unrepaired Data with This Tolerance

		full truth									sure truth								
relation	cand.	true	price	Cert3NF	LLM	w/o adv.	EC² ord.	post.	EC²	θ=0.1	true	price	Cert3NF	LLM	w/o adv.	EC² ord.	post.	EC²	θ=0.1
tpcc-orderline (A)	12,130	67	77	(115†)	(142†)	(252†)	(261†)	–	–	(114)	32	70	(120)	95	(168)	(189)	–	–	(121)
tpcc-stock (A)	238	16	33	48	45	59	53	–	–	48	16	33	34	34	59	63	–	–	34
tpcc-customer (A)	272	12	32	41	42	54	69	–	–	41	12	32	41	40	54	63	–	–	41
tweets (A)	10	2	6	7	8	8	10	10	10	7	2	6	7	9	8	10	10	10	7
syn-encounters (A)	635	13	32	51	39	(105)	71	–	–	54	13	32	48	40	(105)	87	–	–	51
syn-claims (A)	201	7	25	39	42	53	60	–	–	45	7	25	37	43	53	57	100	95	43
syn-trans (A)	244	5	27	39	39	57	52	75	90	40	5	27	40	38	57	54	65	70	41
companies	994	0	22	22	22	22	22	–	–	25	0	22	22	22	22	22	–	–	25
fec-cand	718	11	22	35	30	66	33	–	–	43	11	22	29	29	66	47	–	–	36
fec-comm	868	8	20	23	22	62	32	–	–	30	8	20	22	22	62	39	–	–	29
fec-link	657	20	30	41	41	87	65	–	–	55	18	29	37	36	88	54	–	–	50
gleif	4,008	0	24	24	24	24	24	–	–	25	0	24	24	24	24	24	–	–	25
hgnc	761	327	227	(691†)	(723†)	–	–	–	–	(311)	23	23	25	26	27	43	–	–	26
nppes	327	0	8	8	9	8	8	–	–	24	0	8	8	8	8	8	100	–	24
onspd	311	83	53	(194†)	(182†)	(569†)	(356†)	–	–	(157)	20	27	56	54	(125)	(101)	–	–	93
pay-general	741	11	26	36	34	60	74	–	–	70	10	25	34	33	56	64	–	–	46
pay-profile	122	3	18	20	20	30	29	65	65	29	3	18	20	20	30	30	50	50	29
air-airports	557	62	33	72†	69†	(130†)	73†	–	–	100†	62	31	61	63	(115)	61	–	–	63
air-freq	1,574	45	46	(124†)	(118†)	(200†)	(133†)	–	–	(252†)	45	39	49	47	(130)	58	–	–	52
air-navaids	5,271	29	30	39	44	72	63	–	–	46	29	30	38	35	72	49	–	–	42
se-postlinks	1,237	19	43	56	45	(115)	98	–	–	56	19	43	51	46	(115)	(102)	90	95	51
se-posts	1,053	11	35	41	38	100	83	–	100	42	11	35	40	39	100	77	98	93	41
se-users	18	0	8	8	8	8	8	10	10	8	0	8	8	8	8	9	10	10	8
within 100				19/23	19/23	16/23	19/23	4/23	5/23	19/23			22/23	23/23	17/23	20/23	8/23	7/23	22/23
within 50				16/23	18/23	6/23	8/23	2/23	2/23	14/23			19/23	20/23	7/23	10/23	3/23	3/23	17/23

mechanisms of Thm. 3.5 thus dominate on different data: fabricated keys cause most lossy designs on random schemata (last row of Tab. 1), and path loss causes every lossy design on TPC-H. In summary, a single false or missing FD damages the structure of most synthesized designs, so a wrong estimate costs a migration of the data, not just a worse design. Hence, both kinds of error have to be resolved before deployment, and certification has to settle almost every plausible accidental FD.

6.3 Q2: What Does a Certified Design Cost?

Since almost every accidental FD must be settled, we ask whether an automatic method can do so, and otherwise what a certificate costs a designer (Tab. 3).

Automatic methods. Tab. 4 shows that no fully automatic method reliably obtains a good design: by the median of its seeds, only the normalizer obtains any, on three of the 23 confirmatory relations under the full truth and two under the sure one. Indeed, a good design has to preserve every true FD and avoid the damage of false ones, and each method fails at least one of the two. The mined FDs often contain thousands of accidental ones, the model refutes up to four fifths of the true candidates of a relation or omits attributes and rules in the designs it writes, and the normalizer infers only part of the rules. Graded by the damage of Sec. 3, the designs of these methods are lossless on at most five (full) and three (sure) of the 23 confirmatory relations. Even a designer who judges every candidate obtains a good design only where every true FD is a minimal FD of the data and the data satisfy Σ, which fails on six development relations and on five confirmatory ones. Hence, the designer is not optional and the candidates are not enough, and the question becomes what a designer has to answer.

Table 4: Good Designs, by the Median of Three Seeds, and Macro F1 of the Selected FDs, Averaged over the Seeds; EC² and EC² Order (Cert3NF with the Order of Questions of EC²) Ran on the Confirmatory Relations Only

	good designs				F1			
	dev.		conf.		dev.		conf.	
method	full	sure	full	sure	full	sure	full	sure
mined FDs	0/35	0/35	–	–	.29	.18	–	–
trust	0/35	0/35	–	–	.33	.24	–	–
classifier	0/35	0/35	–	–	.43	.54	–	–
posterior alone	0/35	0/35	–	–	.42	.34	–	–
rules	0/35	0/35	–	–	.38	.32	–	–
LLM design	0/35	0/35	–	–	.19	.24	–	–
normalizer	1/35	3/35	3/23	2/23	.36	.43	.33	.35
every candidate	23/35	30/35	–	–	1.00	1.00	–	–
posterior certificate	18/35	24/35	16/23	19/23	.87	.94	.82	.91
EC²	–	–	16/23	17/23	–	–	.79	.86
EC² order	–	–	18/23	22/23	–	–	.93	.99
Cert3NF	26/35	34/35	18/23	23/23	.98	1.00	.95	1.00

Certificates within the budget. Tab. 3 shows that a designer who answers a hundred questions certifies the design of almost every relation: Cert3NF certifies 19 of the 23 confirmatory relations under the full truth and 22 under the sure truth. Every certificate

is correct where the data satisfy the documented FDs (V2 holds), and the answers stay close to the bound for every method. The certified runs pay a median of 1.30× (full) and 1.22× (sure) the price of Thm. 4.4. On the development relations (Tab. 9), Cert3NF certifies 31 of 35 (full) and 34 of 35 (sure) at a median of 1.34× and 1.30× the price.

Interactive baselines. The framework of Cert3NF, more than the choice of questions, makes a certificate affordable. The posterior certificate and EC^2 ask about candidates alone, and they certify 4 and 5 of the 23 relations within the budget under the full truth, and 8 and 7 under the sure truth (Tab. 3). The posterior certificate also trusts the candidates and the model, and errs where a true FD is not minimal on the data or the model refutes a true FD: on the development relations, it is wrong on 3 of the 17 relations it certifies under the full truth and on 4 of 19 under the sure truth (Tab. 9), while Cert3NF is never wrong. On the confirmatory relations, every documented FD that holds on the data is minimal there, so that no certificate of any method is wrong, and the methods differ in their designs. At the end of the budget, Cert3NF has more good designs than the posterior certificate, 18 against 16 (full) and 23 against 19 (sure), and at least as many on each class (V5 holds). Indeed, a no to a maximal set settles every candidate that leads from inside it to an attribute outside it, while the posterior certificate asks about them one by one.

With the order of EC^2, our framework still certifies most relations, 19 (full) and 20 (sure) of the 23 within the budget, with exact certificates. Hence, the questions about maximal sets and the stopping rule, not the order, lift EC^2 from 5 and 7 relations to 19 and 20. The order of Cert3NF still pays, and most where answers are scarce. Fig. 2 shows that within 50 answers Cert3NF certifies 16 (full) and 19 (sure) of the 23 relations, against 8 and 10 with the EC^2 order (last row of Tab. 3). Without a budget, Cert3NF spends 897 and 222 answers above the price, against 1,800 and 682, and it needs a median of 0.70× and 0.65× the answers of the EC^2 order (V3 holds). Indeed, the EC^2 order loses most on finding the FD behind a yes, 1,295 against 862 answers (full) and 533 against 117 (sure), since it starts QuickXplain at once where Cert3NF first tries a confirmed left-hand side or its likeliest candidate.

Designs without a certificate. Where the budget runs out before the certificate, the design of Cert3NF is still lossless, since it rests on confirmed FDs alone. In contrast, the posterior certificate and EC^2 rely on unconfirmed FDs where they stop, and on the confirmatory relations only 14 of their 19 and 12 of their 18 uncertified relations get a lossless design under the full truth. On these relations, the uncertified designs of Cert3NF are also certified in parts: 102 of their 117 schemata under the full truth and all 21 under the sure truth are certified on their own (Prop. 5.4).

Without a budget. Fig. 2 shows that only the budget stops the uncertified runs: without it, Cert3NF certifies every relation (parentheses in Tab. 3 and Tab. 9), 56 (full) and all 58 (sure) within 200 answers, which take about an hour and a half at the pace of the experts in our study (Sec. 6.5). The certified runs then pay a median of 1.37× and 1.26× the price on the confirmatory relations and 1.36× and 1.28× over all relations (V4 holds). Only two of the 58 relations exceed 3× the price, both under the full truth, hgnc and onspd, where the data violate documented FDs and the designer

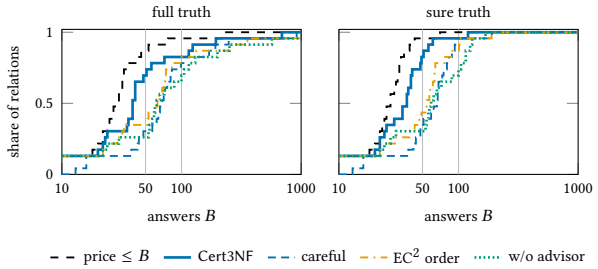


Figure 2: Share of the 23 Confirmatory Relations Certified within B Answers, and Share Whose Price, a Bound for Every Method, Is at Most B

Table 5: Answers until the Certificate on Synthetic Relations without a Budget (Medians over Three Seeds; Share: of the Other Attributes on the Entity Side; Ratio: Answers to Price)

					Cert3NF		w/o advisor
attr.	share	cand.	true	price	answers	ratio	answers
10	0.6	78	5	17	20	1.18	38
20	0.6	16,294	19	42	46	1.11	154
30	0.2	554,220	9	59	64	1.07	113
30	0.4	806,080	20	63	69	1.08	153
30	0.6	145,161	40	61	72	1.18	256
30	0.8	284,217	53	61	73	1.18	268

confirms many FDs with larger left-hand sides in place of each of them.

Attribute exploration. The closest classical method has an expert answer whether an implication holds or construct a counterexample [15, 16]. It determines Σ as exactly as a certificate does, but at several times the answers (Appendix B). With the smallest counterexamples, it does not finish within two hours on eight (full) and nine (sure) confirmatory relations. With the largest, which are maximal sets, it poses a median of 22 implications, fewer than the price, yet these amount to 10.6× (full) and 11.0× (sure) the price in yes and no answers, and to 7.0× and 8.6× those of Cert3NF without a budget. Indeed, most answers of exploration grow the counterexamples, whereas the designer of Cert3NF only judges the maximal sets that the confirmed FDs determine.

Scalability. Tab. 5 shows that the domain and the data, not the number of candidates, set the cost of a certificate. On synthetic snowflake joins whose true FDs hold by construction (Appendix B), the candidates grow from 78 at 10 attributes to about a million on the largest run at 30. Yet the price stays below 73 on every run, and Cert3NF certifies every design correctly at a median of 1.15× the price and 1.30× at most. Indeed, the accidental candidates multiply with the attributes, but the maximal sets settle them in groups. Hence, the answers follow the price (Spearman's rank correlation 0.79 over the 18 relations) and the true FDs (0.82) rather than the candidates (0.49), which confirms the criterion we fixed before generating the data. Without an advisor, Cert3NF needs up to 3.7× as many answers (last column), so the statistics alone save most of the cost where the names of the attributes carry no meaning.

Cert3NF takes less than a second of computation per answer in the median, also with a million candidates, and 3 seconds at most.

Violated FDs. A certificate assumes that the data satisfy Σ, and the tolerance of Prop. 5.6 keeps it exact on data that violate true FDs in a few records. Before all runs, a greedy repair removed records until the sure FDs hold (Appendix B), and the columns $\theta = 0.1$ of Tab. 3 run on the data before the repair. Of the five † relations, only air-airports and air-freq keep their †, since their documented rules fail on 39% and 29% of the records, and every other certificate is correct. Without a tolerance, every certificate on the three relations that the repair changed misses a documented FD under either truth, while $\theta = 0.01$ already makes them all good. The tolerance also saves answers, as on hgnc, where it halves them under the full truth, since the designer no longer confirms many FDs with larger left-hand sides in place of each FD that the data violate. Hence, $\theta = 0.1$, the largest of the three tolerances fixed before these runs, is a good default, which costs about 4% more answers in the median on data that satisfy the sure truth.

6.4 Q3: What Does an Advisor Buy?

Since the certificate does not depend on the advisor, we ask what the advisor contributes and what a wrong one costs.

The statistics. The statistics of the data alone buy most of the savings. Without an advisor, Cert3NF certifies 16 (full) and 17 (sure) of the 23 confirmatory relations within the budget, against 19 and 22 with it, and 13 and 18 of the 35 development relations, against 31 and 34 (Fig. 2). Without a budget, it spends 2,264 and 923 answers above the price on the confirmatory relations, against 897 and 222. Indeed, QuickXplain has to find every revealed FD one attribute at a time, where the advisor guesses it in one.

The order. Choosing the next question after every answer also pays. Even without a model, Cert3NF needs a median of 0.89× (full) and 0.85× (sure) the answers of the two phases with a model, and 1,774 against 2,322 and 851 against 1,198 in total (V1 holds). Indeed, the two phases ask first about the maximal sets most likely to hold and pay for the FD behind every yes, while Cert3NF first asks about the sets that a no is likely to settle, and asks about a likely FD directly, at one answer instead of two. Among its rules, the latent class of a left-hand side saves most, 11% (full) and 18% (sure) of the answers, and the first guesses and the reuse of left-hand sides save 7% each under the full truth, where more maximal sets hold.

The model. On top of the statistics, the model adds little. With it, Cert3NF certifies 19 (full) and 23 (sure) of the 23 relations within the budget, against 19 and 22 without it, and without a budget it needs 1,786 and 811 answers, against 1,774 and 851. No certificate is wrong with or without it, so the model stays optional, which also spares a deployment its cost. With the same model, the EC^2 order gets worse: it spends 2,865 and 1,488 answers above the price, and Cert3NF needs a median of 0.59× and 0.52× its answers. Indeed, its sampler weighs the samples by every answer of the model at once, so that a few samples carry the weight, while Cert3NF asks it only where its answer can change the next question.

A second model. A stronger model does not change this, since two models err on the same candidates. With gpt-oss-20b, which errs on 29% of the confirmatory candidates while Qwen errs on 41%, Cert3NF certifies 19 relations under the full truth and 23 under the sure truth within the budget, against 19 and 22 without a model. Indeed, where one model errs, the odds that the other errs are 5.4× higher, so a stronger model lowers the error rate but leaves the hard candidates to the designer.

A wrong advisor. Prop. 5.2 bounds the damage of a wrong advisor, and it stays small in practice. Cert3NF certifies at least as many relations as without an advisor, and it needs at most b more answers than without one, plus one per maximal set answered yes and per wrong first guess, on all 23 confirmatory relations under both truths (V5 holds).

Table 6: Certificates within the Budget and Wrong Certificates (✗) of a Designer Who Errs with Probability ε, over Relations and Three Seeds (105 Development and 69 Confirmatory Runs per Cell); Repeated: the Designer Repeats a Wrong Answer; ϕ: Correlation of Two Answers to a Question

		development				confirmatory			
		full		sure		full		sure	
ε	mode	cert.	✗	cert.	✗	cert.	✗	cert.	✗
0.02	trusting	73	31	82	28	43	22	53	26
0.02	careful	47	1	56	0	29	0	42	1
0.05	trusting	50	36	54	39	37	26	42	29
0.05	careful	24	1	25	2	16	0	22	1
0.05, repeated	careful	25	11	26	12	19	8	26	13
0.02, $\phi = 0.25$	careful	39	4	51	4	35	3	42	5
0.02, $\phi = 0.5$	careful	39	9	47	6	36	7	48	9
0.05, $\phi = 0.25$	careful	25	13	24	10	13	5	17	5
0.05, $\phi = 0.5$	careful	28	15	29	10	17	5	22	9

6.5 Q4: Does a Certificate Survive a Designer Who Errs?

The designer of (Q2) and (Q3) never errs. We now let the simulated designer answer wrongly with probability $\varepsilon \in \{0.02, 0.05\}$, independently per answer and with three seeds, and compare the careful mode of Cert3NF with a trusting mode that accepts all answers.

Independent errors. Tab. 6 shows that a certificate that trusts every answer breaks under a few wrong ones, while the careful mode keeps the wrong certificates within the bound of Prop. 5.5. Over all relations and both truths, trusting the designer makes 237 of 434 certificates wrong, while answering every premise twice leaves 6 of 261, 1% of the certificates at $\varepsilon = 0.02$ and 5% at $\varepsilon = 0.05$. These are about 1% of the runs, within the 12% and 73% that Prop. 5.5 gives for a budget of 100 answers. The budget also protects the careful mode, since more questions leave more room for two wrong answers: without it, 7% and 42% of the certificates are wrong.

Second answers. The careful mode pays for this correctness in answers: with a designer who never errs, it asks every premise once more and nothing else, so that a certificate costs the answers of the trusting mode plus its premises, between 1.3× and 2× as many and about 1.75× in the median (Fig. 2). Within the budget, it certifies 41 (full) and 49 (sure) of the 58 relations, 18 and 22 of them confirmatory, and without a budget every relation, 56 and 57 within 300 answers.

Table 7: Nine Designers, Three per Condition (E: Experts; P, U: Practitioners with and without Documentation); Block: Fixed Questions about Candidates and Maximal Sets; $\hat{\phi}$: Correlation of Two Answers to a Question

errors (%)	E	P	U	sessions	E	P	U
block candidates	0	3	12	answers per session	55	46	45
block maximal sets	0	8	17	minutes per session	26	35	36
in the sessions	1	10	22	certified, of six	6	3	1
don't know (%)	0	3	7	wrong, full truth	0	1	0
$\hat{\phi}$	0.1	0.7	0.8	wrong, sure truth	1	1	1

Correlated errors. Asking again catches the part of the errors that is independent, and no repetition can catch a designer who repeats a wrong answer (row "repeated" of Tab. 6). Between independent and repeated errors, every question has its own error rate, drawn with mean ε so that two answers to it err together with correlation ϕ (last rows of Tab. 6). Pooled over all relations and both truths, wrong certificates at $\varepsilon = 0.02$ grow from 1% at $\phi = 0$ to 10% at $\phi = 0.25$ and 18% at $\phi = 0.5$, still below the 43% of the trusting mode.

Human designers. Tab. 7 reports a pre-registered study with nine people, in which the experts certified their designs in one session and the practitioners mostly did not (Appendix B). Three database experts, who studied the documentation before each session, and six practitioners, half of whom read it for ten minutes in the session, each worked with the careful mode of Cert3NF, in the two phases, on two of ssb-customer, tpcds-core and abs, within a hundred answers or forty minutes per session. The experts certified all six of their designs, with 55 answers in 26 minutes per session on average. They erred on 1% of their answers, and none of their certificates is wrong under the full truth. The practitioners erred on 22% of their answers and certified one of six designs without the documentation, and erred on 10% and certified three of six with ten minutes of it. Two answers of theirs to the same question err together with correlation $\hat{\phi} = 0.8$ and 0.7, so their errors are beliefs rather than slips. Hence, the careful mode catches slips, and no proof from answers can catch false beliefs: a certificate is as correct as the beliefs of its designer.

7 RELATED WORK

Schema design. 3NF synthesis [7, 10] and its recent optimizations for integrity maintenance and update efficiency [31, 48, 51–53] assume that the given FDs are correct. Wang et al. normalize a mediated schema with probabilistic FDs estimated from web sources [46], and Link and Prade scale 3NF and BCNF to degrees of certainty of FDs [30]. In both, the design is judged against the uncertain input, while we judge it against the unknown true FDs.

Dependency discovery. FD discovery algorithms return all FDs that hold on an instance [23, 38, 39], and Dep-Miner derives them from the agree sets and maximal sets of the data [32]. Wei and Link rank the discovered FDs by the redundancy they cause [47], and Wan and Han discover only the k FDs that cause the most [45]. Others find reliable FDs by statistical tests or models of noise [8, 20, 35, 50], and Normalize chooses among them to decompose a relation into BCNF [40]. These approaches improve the estimate, but leave errors that only a designer can settle.

Confirming dependencies. Armstrong relations satisfy exactly the FDs implied by a given set [4, 6], and Mannila and Räihä let designers confirm FDs by example [36]. Langeveldt and Link show in a user study that they help designers recognize the meaningful FDs [29], and De Marchi and Petit build them from the records of an existing database [37]. The maximal sets we ask about are the agree sets of such relations and the characteristic models of the Horn theory that the FDs form [26]. Our questions about maximal sets thus are those of design by example, but only about the held sets, which the data cannot refute, and Thm. 4.4 shows that no proof needs fewer. Formal concept analysis treats FDs as implications over the pairs of records [5], and its attribute exploration asks an expert whether the implications of a minimal basis hold, where the expert refutes an implication by a counterexample [15, 16]. Seeded with the agree sets, it determines Σ as a certificate does, but its expert constructs the counterexamples and its questions follow the basis rather than the design (Sec. 6.3). Learning such a theory [2, 14] or a minimal cover of FDs [22] needs, besides membership or entailment queries, equivalence queries, which return counterexamples, or closure queries, which return a whole closed set [3], whereas our designer only answers yes or no. By Thm. 4.4(d), our answers form a teaching set [17, 21] for Σ^+ among the closed sets of FDs that hold on the data, and by part (c) the price is its smallest size, which, unlike the teaching dimension, depends on the data.

LLMs as oracle or designer. LLMs predict data correlations from column names [44], and have been asked instead of people as the oracle that decides which statistically strong FDs are meaningful [49], generates ontological FDs for data cleaning [9], or detects foreign keys [13, 42]. They have also been made the designer that normalizes schemata [24], where inferring the FDs is the main bottleneck [28], or that drafts domain models, also with a human who answers questions [11, 41]. In both roles nothing bounds the risk that the result is wrong, and in our experiments neither role reliably obtains a good design on a confirmatory relation. Our work is the first to price the certificate of a design against the unknown true FDs, and to let the data and an unreliable advisor lower its cost but not its correctness.

8 CONCLUSION AND FUTURE WORK

Starting from the observation that estimated FD sets are wrong, we have shown how a designer certifies a schema design against the true FDs at a price set by the domain and the data rather than by the candidates. A confirmed cover and refuted maximal sets settle the candidates in groups, and a smallest cover with the held sets is a shortest proof from yes and no answers. When the data form an Armstrong relation for the true FDs, the certificate is a confirmed smallest cover, the input of classical design.

Future work will address i) designers who answer by example rather than by yes or no, as with Armstrong relations [36], ii) the certification of several relations linked by foreign keys, iii) certificates for BCNF, and iv) stronger models as advisors. Our results show that the answers of a designer who knows the domain follow the length of the proof, and that advice that may be wrong can only change the order of the questions.

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A PROOFS

We prove the results in the order of the paper. Losslessness is decided by the chase of the tableau of $\mathcal{D}$ [1, 34], which has one row per schema S, with the distinguished symbol of column A for every $A \in S$ and fresh symbols elsewhere. The chase with Σ equates the symbols in column A of rows that agree on X, for $X \to A \in \Sigma$, and prefers the distinguished symbol. The design $\mathcal{D}$ is lossless if and only if some row ends with distinguished symbols only. Several proofs reuse the argument of Lem. 2.1. It chases the row of the schema S that contains a key along a derivation of $S^+_\Sigma = R$ by the FDs of F. When the derivation applies $X \to A$, embedded in S', the rows of S and S' agree on X in distinguished symbols, so the row of S receives the distinguished symbol in column A.

Proof of Lem. 2.3. The first claim holds because Σ_a, the keys and the fixed choices depend only on Σ^+. The union of the output is R: an attribute in no kept schema occurs in no FD of G and so lies in every key, hence no kept schema contains a key, and the added key contains the attribute. The output also embeds G and contains a key, so it is lossless and dependency-preserving by Lem. 2.1. Each kept schema XA is in 3NF: X is a key of XA by the minimality of X, and a non-trivial FD $Z \to B$ within XA either has $B = A$, so that $Z \subseteq X$ and hence $Z = X$, or has $B \in X$, which is prime. The key schema K carries no non-trivial FD, since $K \setminus \{B\}$ would otherwise be a superkey of R. For the last claim, the kept covers are non-redundant subsets of Σ_a by construction. Conversely, a non-redundant cover $G \subseteq \Sigma_a$ is kept by an order that walks $\Sigma_a \setminus G$ first: each of these FDs is implied by $G \subseteq \Sigma'$ and removed, after which $\Sigma' = G$ and no FD of G is redundant. □

Proof of Prop. 3.3. (a) Every relation that satisfies Σ satisfies $\hat{\Sigma}$ and is therefore the join of its projections.

(b) If some declared FD is not in Σ^+, it is in $\hat{\Sigma}^+$, so $\hat{\Sigma} \not\subseteq \Sigma^+$. Conversely, if every declared FD is in Σ^+, preservation gives $\hat{\Sigma} \subseteq (\bigcup_S \hat{\Sigma}[S])^+ \subseteq \Sigma^+$. For the rejection, let $Y \to B \in \hat{\Sigma}[S] \setminus \Sigma^+$ and take two tuples that agree exactly on Y^+_Σ. They satisfy Σ, since the set on which they agree is closed, but they violate $Y \to B$, and so does their projection onto $S \supseteq YB$.

(c) Let $\sigma \in \Sigma^+ \setminus \hat{\Sigma}^+$ and let r be a two-tuple relation that satisfies $\hat{\Sigma}$ and violates σ. The projections of r satisfy the declared constraints. If they were the projections of some s that satisfies Σ, then s would satisfy $\hat{\Sigma}$, and losslessness with respect to $\hat{\Sigma}$ would give $s = \bowtie_S s[S] = \bowtie_S r[S] = r$, which violates σ. □

Proof of Prop. 3.4. We have

$$\Sigma_1 \subseteq (\textstyle\bigcup_S \Sigma_1[S])^+ = (\bigcup_S \Sigma_2[S])^+ \subseteq {\Sigma_2}^+,$$

and $\Sigma_2 \subseteq {\Sigma_1}^+$ holds symmetrically. □

The proof of Thm. 3.5 rests on two lemmas, which describe what one more FD does to closures and to the chase.

Lemma A.1 (One more FD). *Let $\hat{\Sigma} = \Sigma \cup \{\sigma\}$. Then $Z^+_{\hat{\Sigma}} = Z^+_\Sigma$ if $X \not\subseteq Z^+_\Sigma$, and $Z^+_{\hat{\Sigma}} = (Za)^+_\Sigma$ otherwise.*

Proof. Let Z' denote the right-hand side, which is contained in $Z^+_{\hat{\Sigma}}$ since $\hat{\Sigma} \supseteq \Sigma$ and $a \in Z^+_{\hat{\Sigma}}$ whenever $X \subseteq Z^+_\Sigma$. Conversely, Z' contains Z, is closed under Σ, and is closed under σ, because $X \not\subseteq Z'$ in the first case and $a \in Z'$ in the second, so $Z^+_{\hat{\Sigma}} \subseteq Z'$. □

Let $\hat{\Sigma}$ and $\hat{\mathcal{D}}$ be as in Thm. 3.5, let $\hat{G}$ be the cover that $\mathcal{A}_\pi$ keeps on $\hat{\Sigma}$, and let W_S be the set of columns in which the row of schema S holds distinguished symbols after the chase of the tableau of $\hat{\mathcal{D}}$ with Σ.

Lemma A.2 (Rows and superkeys). *(a) If $X \not\subseteq S^+_\Sigma$ for a schema $S \in \hat{\mathcal{D}}$, then $W_S = S^+_\Sigma = S^+_{\hat{\Sigma}}$.*

(b) If some schema of $\hat{\mathcal{D}}$ is a superkey with respect to $\Sigma[\hat{\mathcal{D}}]$, then $\hat{\mathcal{D}}$ is lossless with respect to Σ. If $\hat{\mathcal{D}}$ is lossless with respect to Σ, then some schema of $\hat{\mathcal{D}}$ is a superkey with respect to Σ. The two conditions coincide when $\hat{\mathcal{D}}$ preserves Σ.

Proof. (a) For the upper bound $W_S \subseteq S^+_\Sigma$, let t and s be two tuples that agree exactly on S^+_Σ, so that $\{t, s\}$ satisfies Σ. Mapping the row of S to s, every other row to t, and each distinguished symbol to the value of t in its column is a homomorphism from the tableau into $\{t, s\}$, because s and t agree on S. Every chase step equates two symbols that this homomorphism already maps to the same value, so the chased tableau maps into $\{t, s\}$ as well. A column in which the row of S is distinguished is therefore one in which s and t agree, that is, a column in S^+_Σ. For the lower bound, $S^+_{\hat{\Sigma}} = S^+_\Sigma$ by Lem. A.1, and $S^+_{\hat{\Sigma}}$ is derived from S by the FDs of $\hat{G}$. Every FD $Z \to C$ used in this derivation has $Z \subseteq S^+_\Sigma$, so $X \not\subseteq Z^+_\Sigma$, and it is in Σ^+ by Lem. A.1, and as an FD of $\hat{G}$ it is embedded in some schema of $\hat{\mathcal{D}}$. Following the derivation as in the proof of Lem. 2.1, the row of S receives the distinguished symbol in every column of $S^+_{\hat{\Sigma}}$.

(b) The first statement follows from the argument of Lem. 2.1, since every FD of $\Sigma[\hat{\mathcal{D}}]$ is true and embedded in a schema of $\hat{\mathcal{D}}$. For the second, a row of distinguished symbols only belongs to a schema S with $R = W_S \subseteq S^+_\Sigma$ by the upper bound in (a). If $\hat{\mathcal{D}}$ preserves Σ, then $(\Sigma[\hat{\mathcal{D}}])^+ = \Sigma^+$, and the two conditions coincide. □

Proof of Thm. 3.5. (a) Let $S^+_{\hat{\Sigma}} = R$. If $a \in S^+_\Sigma$, then $S^+_\Sigma = S^+_{\hat{\Sigma}} = R$ by Lem. A.1, whether or not $X \subseteq S^+_\Sigma$, and conversely $S^+_\Sigma = R$ contains a. A schema with $S^+_{\hat{\Sigma}} \neq R$ is not a superkey under Σ either, since $S^+_\Sigma \subseteq S^+_{\hat{\Sigma}}$. This proves the equivalence, and lossiness follows from Lem. A.2(b). Finally, $\mathcal{A}_\pi$ ensures that some schema contains a minimal key K of $(R, \hat{\Sigma})$, and if $K^+_\Sigma = R$, that schema is a superkey under Σ.

(b) The first statement is the contrapositive of the last sentence of Lem. A.2(b). For the second, suppose that $Y \to B \in \Sigma^+$ is not implied by $\Sigma[\hat{\mathcal{D}}]$ while $X \not\subseteq Y^+_\Sigma$. Then $Y^+_{\hat{\Sigma}} = Y^+_\Sigma$ by Lem. A.1, and as in the proof of Lem. A.2(a) the derivation of B from Y by $\hat{G}$ only uses FDs of Σ^+ that are embedded in schemata S' of $\hat{\mathcal{D}}$. Each of them is implied by $\Sigma[S']$, so $\Sigma[\hat{\mathcal{D}}]$ implies $Y \to B$, a contradiction. □

Proof of Thm. 4.2. Suppose that $M \to A \notin \Sigma^+$ for every A and every $M \in \max(H, A)$, and let $X \to A \in \Sigma^+$. If H did not imply $X \to A$, then $X \subseteq M$ for some $M \in \max(H, A)$, and $M \to A$ would follow from $X \to A$ by augmentation, a contradiction. Hence, $\Sigma^+ \subseteq H^+$, and $H \subseteq \Sigma^+$ gives equality. Conversely, if $H^+ = \Sigma^+$ and $M \in \max(H, A)$, then $M \to A \notin H^+ = \Sigma^+$. Finally, validity depends only on the closure of the FD set by its definition, and so does $\mathcal{A}$ by Lem. 2.3, so a design that is good for H is good for Σ. The atomic closure also depends only on the closure, so $H[S] = \Sigma[S]$ for every schema S. □

Proof of Thm. 4.4. (a) If M contains the left-hand side of a candidate $X \to A$, then $M \to A$ holds on r by augmentation, since every candidate holds on r. Conversely, if $M \to A$ holds on r, let $X \subseteq M$ be minimal with $X \to A$ holding on r. Since $H \supseteq \Sigma_0$ does not imply $M \to A$, the FD $X \to A$ is not implied by Σ_0, so it is a minimal FD of r outside ${\Sigma_0}^+$, that is, a candidate.

(b) After the yes answers, $H = \Sigma_0 \cup F$, so $H^+ = \Sigma^+$ and $\max(H, A) = \max(\Sigma, A)$ for every A. A maximal set on which $M \to A$ fails on r is refuted by two tuples of r, which satisfies Σ, so $M \to A \notin \Sigma^+$ without a question, and the remaining ones are the held sets. Let M be held, let $M \to A$ be its no answer, and let $M \in \max(\Sigma, A')$ with $M \to A'$ holding on r. Then $M = M_H^+$, since M is closed under Σ, and $A \in (MA')_H^+$, since $(MA')_\Sigma^+$ is a closed set that contains M and $A' \notin M$, so it contains A by the maximality of M for A. Hence, the no answer to $M \to A$ refutes $M \to A'$, every maximal set of H is false, and Thm. 4.2 certifies every design that is good for H, that is, for Σ.

(c) Consider a set of answers that certifies a design. Every maximal set of H is false, so $(\Sigma_0 \cup \Sigma_y)^+ = H^+ = \Sigma^+$ by Thm. 4.2. The held sets are then the $M \in \max(H, A)$ for which $M \to A$ holds on r, each refuted by a no answer to some $Z \to B$ with $M \subseteq Z_H^+$ and $B \in (ZA)_H^+$. If $A \in Z_H^+$, then $B \in (ZA)_H^+ = Z_H^+$ and $Z \to B \in H^+ = \Sigma^+$, which contradicts the truthful no. Hence, Z_H^+ is a closed set that contains M and not A, and $M = Z_H^+$ by the maximality of M. Thus a no answer to $Z \to B$ refutes the maximal sets of the one held set Z_H^+, and a certificate has at least one no answer per held set.

(d) Let a set of answers certify a design that is good for Σ, so that $H^+ = \Sigma^+$ as in (c), and let Σ' be a closed set of FDs that holds on r, contains H, and contains no FD answered no. Then $\Sigma^+ \subseteq \Sigma'$. If $X \to A \in \Sigma' \setminus \Sigma^+$, then $X \subseteq M$ for some $M \in \max(\Sigma, A) = \max(H, A)$, and $M \to A \in \Sigma'$ holds on r, so a no answer to some $Z \to B$ refutes it. That is, $M \subseteq Z_H^+$ and $B \in (ZA)_H^+$, so $Z \to B \in \Sigma'$, a contradiction. Hence, $\Sigma' = \Sigma^+$. Conversely, let Σ^+ be the only such set. The closure of H is one, since r satisfies Σ and $H^+ \subseteq \Sigma^+$ contains no FD answered no, so $H^+ = \Sigma^+$. For $M \in \max(H, A)$ with $M \to A$ holding on r, the closure of $H \cup \{M \to A\}$ holds on r and differs from Σ^+, so it contains an FD $Z \to B$ answered no. Since $B \notin Z_H^+$, this requires $M \subseteq Z_H^+$ and $B \in (ZA)_H^+$, so the no answer refutes $M \to A$. Hence, every maximal set of H that holds on r is refuted, and the answers certify every design that is good for H, that is, for Σ. □

Proof of Prop. 5.1. A no answer to a candidate $X \to A$ with $X_H^+ = M$ is a no answer to $M \to A$, since $H \subseteq \Sigma^+$ and $X \subseteq M$. Let Cert3NF ask about $M \to A$ with $M \in \max(H, A)$ and receive no. Since $H \subseteq \Sigma^+$ and $M \to A \notin \Sigma^+$, $M \subseteq M'$ for some $M' \in \max(\Sigma, A)$; M' is closed under Σ, hence under H, and does not contain A, so $M = M'$ by the maximality of M in $\max(H, A)$. Cert3NF asks only about maximal sets that contain the left-hand side of a candidate, so $M \to A$ holds on r by Thm. 4.4(a), and M is held. For every attribute $A' \notin M$, $(MA')_H^+$ is a closed set that strictly contains M, so it contains A by the maximality of M, and it only grows with H. Hence, the no answer to $M \to A$ refutes $M \to A'$ under every later H, so no later question concerns M. □

Proof of Prop. 5.2. Every candidate holds on r and is minimal there, so a true candidate has a minimal left-hand side under Σ as well, since a smaller true left-hand side would hold on r. A yes answer to a candidate therefore puts into H a true candidate with a minimal left-hand side that H did not imply, and QuickXplain returns a minimal $X \subseteq M$ with $X \to A \in \Sigma^+$, which H did not imply at the yes to $M \to A$, since M is closed under H and does not contain A. Once $X \to A$ is in H, no later maximal set for A contains X, so the same FD is not found twice, and no later question concerns a candidate in H; hence the d FDs are different, and Σ_0 implies none of them. The questions are of five kinds. Believed FDs receive at most b no answers and one yes per FD. Maximal sets and the candidates asked in their place receive at most m_r no answers by Prop. 5.1, and one yes per FD. After a yes to $M \to A$, there are at most g tries, among them a confirmed left-hand side that narrows the search, and $J(|X|, |M|)$ questions of QuickXplain if the tries miss. A first guess before a maximal set receives one yes per FD, and a miss counts among the g tries if the set holds. If the set is refuted, the miss is charged to that set, which gives at most m_r such misses. They are also at most d: Cert3NF swaps only if $p(1 + h') > 1$, and since $h' < 1$, this requires $p > 1/2$, that is, more maximal sets that held than refuted ones so far, so the t-th such miss follows at least t sets that held, each with an FD of its own. Summing gives the bound. The count d can be exponential in the size of a smallest cover: if Σ consists of $A_i \to B_i$ and $B_i \to A_i$ for $i = 1, \ldots, n$ and of $A_1 \cdots A_n \to C$, a run that settles C before the equivalences confirms one FD for each of the 2^n ways to pick A_i or B_i. □

Proof of Prop. 5.3. Let Σ' consist of the FDs $Y \to A$ with $Y \subseteq M$ and $|Y| = k$, and let r be an Armstrong relation for Σ' [6], on which an FD holds exactly when Σ' implies it. For a k-subset X of M, $\Sigma = \{X \to A\}$ holds on r, and a smallest F has one FD. The maximal sets of Σ are $M \setminus \{x\}$ for A with $x \in X$, and $R \setminus \{B\}$ for $B \in M$. Since $m - 1 \geq k$, Σ' implies $M \setminus \{x\} \to A$, while Σ' implies no FD $R \setminus \{B\} \to B$. Hence, the held sets are the k sets $M \setminus \{x\}$, and the price is $k + 1$. A method chooses each question from the earlier answers, so its runs form a binary tree, and two sets $X \neq X'$ whose runs end in the same leaf receive the same answers. These answers are truthful for both $\{X \to A\}$ and $\{X' \to A\}$, which hold on r and have different closures, so by Thm. 4.4(d) they certify a design for at most one of them. Hence, the tree has at least $\binom{m}{k}$ leaves and a depth of at least $\log_2 \binom{m}{k} \geq k \log_2(m/k)$. □

Proof of Prop. 5.4. For $T \subseteq S$, $T_{H[S]}^+ = T_H^+ \cap S$, since $H[S]$ is a cover of the FDs of H^+ within S. Hence, every $T \subseteq S$ with $A \notin T_H^+$ lies within a member of $\max(H[S], A)$. Let $X \to A \in \Sigma^+$ with $XA \subseteq S$. If H did not imply it, then $X \subseteq T$ for some $T \in \max(H[S], A)$, and $T \to A$ would follow from $X \to A$ by augmentation, a contradiction. With $H \subseteq \Sigma^+$, the two closures contain the same FDs over S. Whether S is in 3NF depends on the FDs over S alone, so S is in 3NF with respect to Σ as it is with respect to H by Lem. 2.3, and $\Sigma[S] = H[S]$, since both consist of the FDs over S that are minimal in the common closure. □

The careful mode of Sec. 5 settles conflicting answers as follows. A net no to an FD that H implies is a *conflict*. It is settled by asking again the question whose majority is smallest, among the no and a

Table 8: The Relations with Data: Attributes, Rows, Candidates, and True Candidates under the Full/Sure Truth; (A) Marks Class A

development										confirmatory				
relation	$\|R\|$	rows	cand.	true	relation	$\|R\|$	rows	cand.	true	relation	$\|R\|$	rows	cand.	true
food	17	312,847	78	31/7	sec	25	999,991	428	16/15	tpcc-orderline (A)	24	300,040	12,130	67/32
hospital	16	1,000	145	54/22	scorecard	25	160,030	5,169	64/33	tpcc-stock (A)	17	100,000	238	16/16
pricepaid	15	926,485	34	13/4	epa	23	381,023	1,895	25/13	tpcc-customer (A)	14	30,000	272	12/12
ssb-customer	8	3,000	24	12/10	pums	25	393,725	2,056	820/8	tweets (A)	5	20,000	10	2/2
ssb-date	17	2,557	209	205/113	chicago	22	259,629	314	21/13	syn-encounters (A)	24	80,771	635	13/13
ssb-lineorder	16	600,572	3,814	13/5	usda	21	644,125	161	69/25	syn-claims (A)	23	153,806	201	7/7
ssb-part	9	20,000	86	3/3	auctionmark	22	34,743	4,243	36/15	syn-trans (A)	22	1,000,000	244	5/5
ssb-supplier	7	200	21	11/9	seats	24	77,399	4,640	292/19	companies	25	849,999	994	0/0
tpcds-core	12	2,685,687	48	23/13	tatp	18	376,152	909	28/28	fec-cand	20	9,796	718	11/11
tpch-lineitem	23	600,572	36,883	10/10	simplewiki	17	999,656	139	36/30	fec-comm	20	20,937	868	8/8
tpch-orders	20	150,000	484	44/44	chembl	25	999,405	385	204/43	fec-link	24	8,619	657	20/18
tpch-partsupp	19	80,000	2,405	25/25	musicbrainz	25	1,000,000	85	71/61	gleif	25	1,000,000	4,008	0/0
tpcds-customer	25	100,000	68,156	44/44	pluto	25	858,284	1,572	202/46	hgnc	24	45,289	761	327/23
ldbc-comment	25	1,000,000	1,784	26/26	fhrs	22	579,855	1,488	25/25	nppes	24	1,000,000	327	0/0
ldbc-study	24	7,949	1,249	39/39	geonames	25	1,000,000	2,900	12/12	onspd	25	998,456	311	83/20
gtfs	24	1,000,000	1,170	23/23	eia-860	25	27,919	1,097	14/14	pay-general	25	999,988	741	11/10
faa	25	317,999	695	14/14	stats19	24	128,272	1,670	42/42	pay-profile	25	1,000,000	122	3/3
abs	17	368,286	78	59/59	ghcn-daily	19	999,369	351	13/13	air-airports	24	86,335	557	62/62
										air-freq	24	30,390	1,574	45/45
										air-navaids	25	10,351	5,271	29/29
										se-postlinks	24	1,581	1,237	19/19
										se-posts	25	15,283	1,053	11/11
										se-users	8	33,730	18	0/0

minimal set of yes answers that imply the FD. A changed net answer rebuilds H, the refutations and the maximal sets, and the loop continues. The certificate is issued when no pair needs a question, no question is unanswered, no conflict is left, and every premise has a majority of at least two answers. A conflict whose questions have all been asked three times ends the run without a certificate.

Proof of Prop. 5.5. If every premise is right, then $H^+ \subseteq \Sigma^+$, since H is generated by Σ_0 and the yes premises, and every maximal set of H is false, since the no premises refute those that hold on r and r, which satisfies Σ, refutes the others; hence $H^+ = \Sigma^+$ by Thm. 4.2, and the design is good for Σ. The certificate is therefore wrong only if some premise is wrong. Think of the answers as drawn in advance, three for every question, each wrong independently with probability ε, of which Cert3NF reads the first one, two or three, since no question receives more than three answers. A wrong premise is a wrong majority of at least two answers, which requires at least two of the three drawn answers of its question to be wrong, with probability $3\varepsilon^2(1-\varepsilon) + \varepsilon^3 = 3\varepsilon^2 - 2\varepsilon^3$. Number the different questions in the order in which they are first asked. Whether a j-th question is asked, and which one, depends only on the answers read before its first answer, which are independent of its own drawn answers. Hence, the probability that a j-th question is asked and two of its drawn answers are wrong is $\Pr[Q_r \geq j](3\varepsilon^2 - 2\varepsilon^3)$, where Q_r is the number of different questions of the run, and the sum over j is $Q(3\varepsilon^2 - 2\varepsilon^3)$. Within a budget of 100 answers, $Q \leq 100$, and $100(3\varepsilon^2 - 2\varepsilon^3) < 300\varepsilon^2$. With an error rate ε_q per question, drawn independently of the other questions, the same argument applies with the probability $3\mathrm{E}[\varepsilon_q^2] - 2\mathrm{E}[\varepsilon_q^3]$ that two of the three drawn answers are wrong. □

Proof of Prop. 5.6. The data refute an FD only if it fails on r up to θ, so the FD is not in Σ^+. For $X \subseteq M$, every group of records that agree on M lies within a group that agrees on X, so the records that remain after the removal for $X \to A$ also satisfy $M \to A$. Hence, $M \to A$ holds up to θ whenever $X \to A$ does, and (a) follows as in the proof of Thm. 4.4 with a minimal $X \subseteq M$ for which $X \to A$ holds up to θ. A set of answers that certifies a design refutes every maximal set of H that holds on r up to θ, and the other maximal sets are refuted by the data and hence false, so the design is good for Σ by Thm. 4.2, which does not refer to the data. The proofs of (b) and (c) use the data only in this way, to refute the maximal sets whose FDs fail on r, and argue about closures otherwise, so they carry over unchanged. Part (d) does not, since the FDs that hold on r up to θ need not be closed under implication. □

B EXPERIMENTAL DETAILS

This appendix gives the protocol of Sec. 6. The artifact holds the implementation, the mined relations with their ground truth and every raw result.

Environment. Synthesis, Cert3NF and the estimators were implemented in Python. The LLM is Qwen3.5-9B in FP8, served by vLLM 0.31.0 on one NVIDIA RTX 4090 in each of three servers, and every other computation ran on the 16 CPU cores and 60 GB of memory of these servers. Times are wall-clock per run and include the advisor, but not the mining of the candidates and the answers of the model, which are computed once beforehand. The second model of (Q3) is gpt-oss-20b in its released MXFP4 weights with medium reasoning effort, served the same way and asked the same prompts, and the error rates of its advisor are estimated from its answers on the development relations, as for Qwen. The model answers one prompt per candidate, which names the attributes, states Σ_0 and shows a few rows, with yes or no and a confidence. Every candidate of every relation is answered once, with a fixed seed per prompt, and Cert3NF reads the answer of a candidate only when it

queries the model. The normalizer runs an evidence, a diagnosis and a generator agent of the same model with up to 32,768 output tokens per call, and a deterministic verifier that checks the design against the inferred FDs and returns it for at most two repairs.

Random schemata and TPC-H. For the random schemata we drew Σ over n attributes with 2 to $n + 1$ FDs, whose left-hand sides have one to three attributes, 300 times for each $n \in \{4, 5, 6\}$ and 150 times for $n = 7$. Into each we injected, one at a time, up to 12 random false FDs $X \to A \notin \Sigma^+$ with $|X| \leq 2$ and every missing FD of the canonical cover. This gave 10,737 false and 2,958 missing FDs for $n \leq 6$, and 1,800 and 655 for $n = 7$. On TPC-H, the true FDs follow from the specification: besides the primary keys, the names of customers, suppliers, nations and regions are keys, the brand determines the manufacturer, the country code of a phone number determines the nation, the part together with the quantity or the extended price determines the other, and the ship date determines the line status. HyFD discovers 3,945 minimal FDs on the eight tables, of which 76 follow from the specification, and we injected each of the 3,869 accidental ones alone.

Development relations. The first block of development relations in Tab. 8 consists of three denormalized joins that cover the tables of TPC-H, the five tables of SSB, a core of TPC-DS, and the real data sets food (Chicago food inspections), hospital and pricepaid (UK land registry). Their ground truth is given by the specification of the benchmark or by the rules that the publisher documents. The relations of the other two blocks were chosen and prepared by the rules below: a customer cluster of TPC-DS, the comments and the study relation of LDBC SNB, the GTFS feed of Auckland Transport, the FAA aircraft registry, the ABS statistical geography, SEC financial statement data sets, the College Scorecard, EPA air quality, the ACS PUMS of California, Chicago crimes and the USDA nutrient database, and the items, flights and subscribers of the AuctionMark, SEATS and TATP benchmarks, the category links of the Simple English Wikipedia, the activities of ChEMBL, the recordings of MusicBrainz, the tax lots of NYC PLUTO, the establishments of the UK Food Hygiene Rating Scheme (fhrs), the places of GeoNames, the generators of EIA-860, the casualties of STATS19 and the daily observations of GHCN-Daily. For every result on the development relations, the learned components of Cert3NF, the prior and the error rates of the model, were trained on the first twelve relations, and for the last twelve on the first 24, leaving the relation under test out when it is one of them. For the confirmatory relations they are trained once on all 36, and the kind of prior and of answer model were chosen by their held-out log-loss before any confirmatory answer was read. The model has no answers for tpcds-customer, whose 68,156 candidates were not all answered, so this relation only trains the prior.

Confirmatory relations. The confirmatory relations were chosen from their documentation alone, before any data were opened, by rules fixed in advance: public, with a specification or data dictionary that states keys, code tables and hierarchies, outside the development relations and the collections used in related work, at least six benchmarks with a published schema (class A) and the rest real data (class B), at most three relations per publisher, each with a root table of its own. Each relation was specified before its data were opened: which tables, which documented foreign keys to join along, which columns to keep, at most 25 columns, at least a thousand rows, and a sample of one million rows with a fixed seed where there were more. Class A consists of the order lines, stock and customers of TPC-C and the tweets of the Twitter benchmark, generated by BenchBase at scale factor one, and the encounters, claims and claim transactions of Synthea. Class B consists of the companies of Companies House, the candidates, committees and linkages of the FEC, the legal entities of GLEIF, the genes of HGNC, the providers of NPPES, the postcodes of ONSPD, the general payments and recipient profiles of CMS Open Payments, the airports, frequencies and navaids of OurAirports, and the post links, posts and users of a Stack Exchange site. The ground truth is the projection of the documented FDs onto the chosen columns, each with a quotation from the documentation, written by a reader of the documentation who did not see the data. An FD that the documentation states explicitly is sure, and one that has to be inferred is not. The data were cleaned as the development relations were, and a greedy repair removed records until the sure FDs hold. It changed only onspd, pay-general and air-navaids and removed at most 1,544 records, which is 6% of the records of air-navaids at most. On the unrepaired data of these three relations, the documented FDs fail on at most 0.15% of the records, except one rule of air-navaids on 6%. Sec. 6.3 also reports Cert3NF on the unrepaired data, with and without the tolerances $\theta \in \{0.001, 0.01, 0.1\}$, which were fixed before any run with $\theta > 0$. After mining, a sample of at most 200 candidates per relation, 4,139 in all, was audited blindly against the documentation by readers who saw neither the ground truth nor any answer of the model. They refuted six documented FDs and added nine candidates that the documentation implies, and inferred FDs were promoted to sure on two relations. Before any method ran, we listed the documented FDs that the data violate, 28 inferred FDs on tpcc-orderline, hgnc, onspd, air-airports and air-freq, and checked that every documented FD that holds on the data is minimal there.

Simulated designer. The documented FDs and the declared key play the designer: a question whether $Z \to A$ holds is answered yes if and only if they imply it, and a question that the data refute is not asked. For (Q4), the simulated designer answers wrongly with probability ε, independently on every answer. Whether an answer is wrong is decided by a hash of the relation, the seed, the question, ε and the number of times the question was asked, so that every method meets the same errors. A designer who *repeats* an error gives the first wrong answer again when asked again.

Rules fixed in advance. Before the first confirmatory relation was evaluated, we wrote the following rules into the design record of the project, judged per ground truth over the 23 relations with a budget of 1,000 answers, with the two classes reported separately. (V1) Paired by relation, the median ratio of the answers of Cert3NF to those of the two phases with the LLM is at most 0.9, and Cert3NF needs fewer answers in total. (V2) Every certificate of Cert3NF is correct: the checker verifies its premises against the ground truth, and its design is good, except on the relations listed before the run where the data violate a documented FD. (V3) The median ratio of the answers of Cert3NF to those of the EC^2 order, with and without the LLM and by the median of three seeds, is at most 1. (V4) The ratio of answers to the price of Thm. 4.4 has a median of at most 1.6 and an upper quartile of at most 2, and no certificate has fewer no answers than m_r. (V5) Cert3NF certifies at least as many relations

Table 9: Answers until the Certificate on the Development Relations, as in Tab. 3; ✗ Marks a Wrong Certificate

		full truth						sure truth					
relation	cand.	true	price	Cert3NF	LLM	w/o adv.	post.	true	price	Cert3NF	LLM	w/o adv.	post.
food	78	31	33	64†	63†	(152†)	75†	7	11	33	33	49	55✗
hospital	145	54	36	67	87	(121)	96✗	22	34	70	66	99	80✗
pricepaid	34	13	15	22†	24†	55†	32†	4	7	12	19	26	33
ssb-customer	24	12	11	15	16	25	18	10	10	11	14	21	19
ssb-date	209	205	23	26	26	(154)	40	113	30	43	42	(118)	70
ssb-lineorder	3,814	13	44	51	53	86	–	5	33	37	37	48	85
ssb-part	86	3	11	12	14	20	35	3	11	14	14	20	40
ssb-supplier	21	11	10	13	16	22	15	9	9	10	13	20	15
tpcds-core	48	23	17	27	26	78	25✗	13	25	67	60	80	23✗
tpch-lineitem	36,883	10	49	54	61	96	59✗	10	49	55	68	96	–
tpch-orders	484	44	32	58	39	(151)	–	44	32	56	42	(151)	–
tpch-partsupp	2,405	25	36	48	44	(108)	–	25	36	52	44	(108)	–
ldbc-comment	1,784	26	40	50	50	(208)	–	26	40	49	49	(208)	–
ldbc-study	1,249	39	46	67	65	(212)	–	39	46	71	65	(212)	–
gtfs	1,170	23	50	79	76	(167)	–	23	50	60	67	(167)	–
faa	695	14	38	55	53	98	80	14	38	55	60	98	95
abs	78	59	27	29	31	(156)	73	59	27	29	30	(156)	67
sec	428	16	35	45	46	(103)	92	15	37	55	55	(109)	100✗
scorecard	5,169	64	62	95	(103)	(161)	–	33	66	86	79	(152)	–
epa	1,895	25	124	(170)	(213)	(202)	–	13	125	(188)	(206)	(194)	–
pums	2,056	820	278	(317†)	(332†)	–	–	8	26	31	31	61	–
chicago	314	21	32	50	50	(115)	100	13	26	40	42	74	95
usda	161	69	41	51	56	(154)	97	25	31	40	40	(125)	70
auctionmark	4,243	36	33	41	41	(114)	–	15	47	54	52	(125)	98
seats	4,640	292	54	(101†)	(102†)	(345†)	–	19	40	57	57	(122)	–
tatp	909	28	38	50	49	81	–	28	38	47	47	81	–
simplewiki	139	36	44	72†	71†	(121†)	–	30	40	48	54	80	100
chembl	385	204	54	100†	(103†)	(674†)	–	43	29	39	44	(156)	–
musicbrainz	85	71	35	47†	50†	(170†)	80†	61	34	46	46	(137)	73
pluto	1,572	202	108	(151†)	(211†)	(488†)	–	46	42	56	55	(111)	–
fhrs	1,488	25	30	34	36	70	–	25	30	34	34	70	–
geonames	2,900	12	43	77	96	(144)	–	12	43	73	75	(144)	–
eia-860	1,097	14	35	43	48	100	–	14	35	40	46	100	–
stats19	1,670	42	33	45	45	79	60	42	33	42	45	79	55
ghcn-daily	351	13	28	37	37	80	98	13	28	36	36	80	80
within 100				31/35	29/35	13/35	17/35			34/35	34/35	18/35	19/35

as without an advisor, needs at most b answers more than it, plus one per maximal set answered yes and per wrong first guess, on at least 90% of the relations, and has at least as many good designs as the posterior certificate at the end of a budget of 100 answers, counted per relation by the median of its three seeds. The rules were judged once. V1, V2, V3 and V5 hold on all relations, on class A and on class B under both truths. V4 holds on all relations and on both classes under both truths, except that the upper quartile on class B under the full truth is 2.03. We find a smallest F exactly, by integer programming over the left-reduced FDs of Σ. Where the data violate a documented FD, F is a smallest cover of the FDs of Σ^+ that hold on the data, and the held sets are the maximal sets of Σ that contain the left-hand side of a false candidate. The automatic methods, the normalizer, the second model of (Q3), attribute exploration and EC^2 ran after these verdicts or apart from them, and no rule depends on them. The posterior certificate and EC^2 draw 80 samples from the posterior, and the posterior certificate accepts a design at $\delta = 0.1$. EC^2 weighs the samples of the posterior equally, assigns each sample to the first design that covers it in a greedy cover of the samples by good designs, and asks per round about the five candidates of largest expected weight of cut edges, with ties broken by the largest marginal.

Nine designers. The experts of the study are database researchers and administrators from outside our group, and its practitioners are graduate students and developers who have taken a database course and designed schemata. Every person worked on two relations in two sessions one to seven days apart, so that in every condition each relation comes first once and second once. A session starts with a fixed block of six questions about candidates, six about maximal sets and six about other subsets, half of them true (on abs only nine questions, all true), and then runs Cert3NF in the careful mode with $\theta = 0$. A question about a maximal

set names the attributes outside it: must any two records that agree on all attributes except B_1, B_2 and A also agree on A? The registered claim that maximal sets are as answerable as candidates, with at most five points more errors at the upper end of the 95% interval, holds for the experts, who erred on neither, and is not established for the practitioners, who erred on 8% (17%) of the maximal sets against 3% (12%) of the candidates with (without) the documentation. The registered claim that errors are mostly independent, with the upper end of the 95% interval of $\hat{\phi}$ below 0.25 for the experts and for the practitioners with the documentation, does not hold for the latter, since $\hat{\phi}$, estimated from the first two answers to each repeated premise, is 0.7 for them against 0.1 for the experts. Every expert passed the required check of at least five right answers to the six candidate questions of a block. An answer of don't know counts towards the budget, is not an error, and leaves its candidate or maximal set undecided. We kept the totals per condition, so that, unlike the registration, we report no intervals clustered by participant and no results per session, and leave out the registered comparison with the errors of the LLMs and the effects of question size and session order.

Attribute exploration. We run exploration by NextClosure [15] on the confirmatory relations, with the attributes in their order in the relation, the agree sets of the data as the known objects, Σ_0 as background knowledge, and the simulated designer as the expert. The classical expert refutes an implication by the smallest counterexample, the closure of its premise under Σ, and we also let it return the largest, a maximal set of Σ that contains the premise. To compare with Cert3NF, an implication becomes one yes or no answer per attribute of its conclusion that the confirmed implications do not yet imply, and the largest counterexample one answer per attribute by which it is grown, unless the data or the confirmed implications decide it. With the smallest counterexamples, exploration poses from 9 to almost three million implications, and eight relations under the full truth and nine under the sure truth do not finish within two hours. With the largest, it poses a median of 22 implications, fewer than the price, since one yes confirms a whole conclusion and one counterexample is a maximal set, and Sec. 6.3 reports the answers they amount to.

Synthetic relations. To see whether the answers follow the price beyond the 25 columns of the real relations, we generated denormalized joins of snowflake schemata whose true FDs hold by construction. The fact key of order and line is declared, the entities form a tree under the fact, each with a key, one to four descriptive attributes and, with probability 0.3, a name that is a second key, and the remaining attributes are measures that only the fact key determines. Cardinalities and domains are drawn log-uniformly, values uniformly, and the attributes are renamed in a random order. One sweep varies the number of attributes at a share of 0.6 of the other attributes on the entity side, and the other varies the share at 30 attributes, each with three seeds of 20,000 rows. Before generating any data, we fixed that Cert3NF and its variant without an advisor certify every design within 1,000 answers (S1), that the median ratio of answers to the price is at most 2 in every configuration (S2), and that Spearman's ρ of the answers is larger with the price than with the number of candidates (S3). All three hold (Tab. 5 in Sec. 6.3). Beyond 30 attributes, the limit lies in mining and in holding the candidates: two of the three relations with 40 attributes are mined within two hours by our miner, with 5.6 and 6.9 million candidates, which take more than the 60 GB of memory of our servers, and none with 50 attributes is.